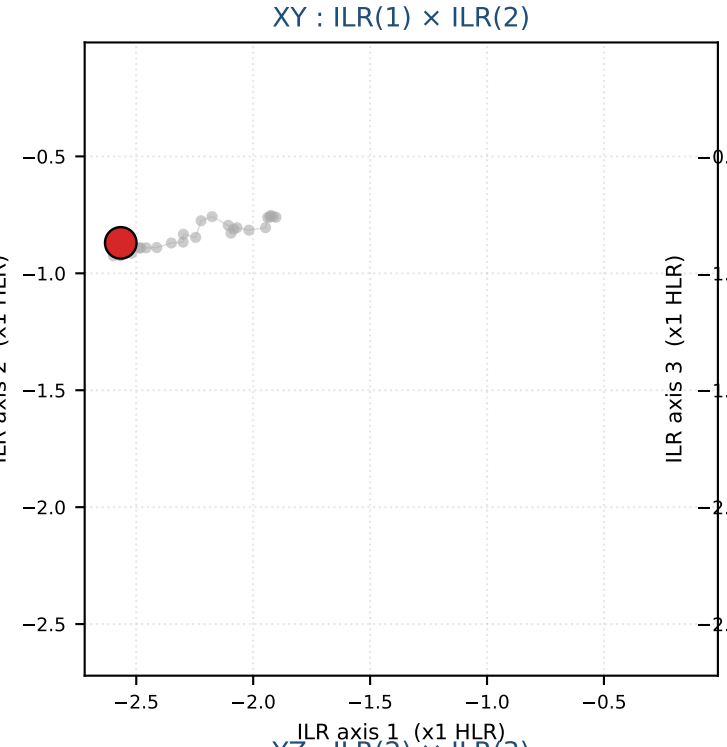
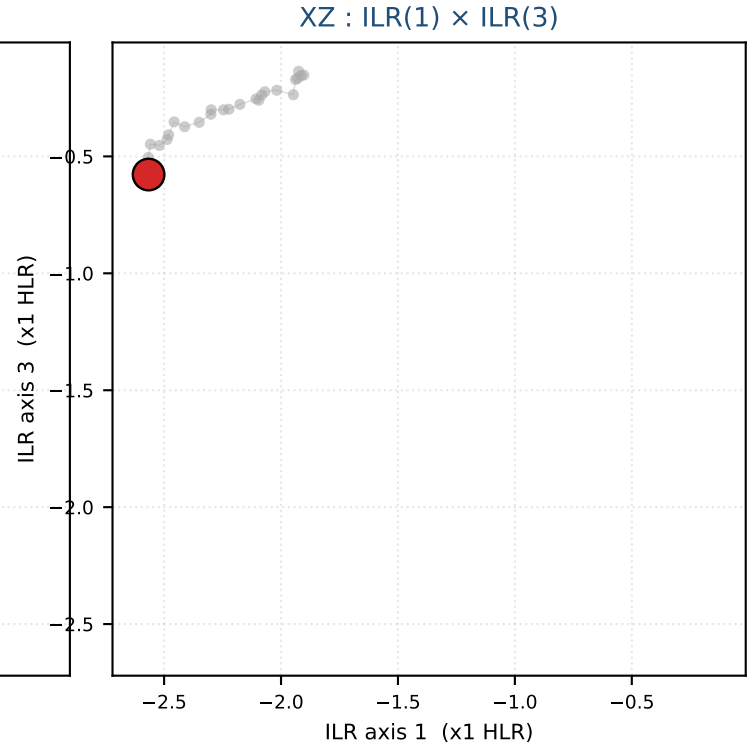
STAGE 1 v4 | t=0 | 2000 | D=9 T=26 | level 1 first-principles

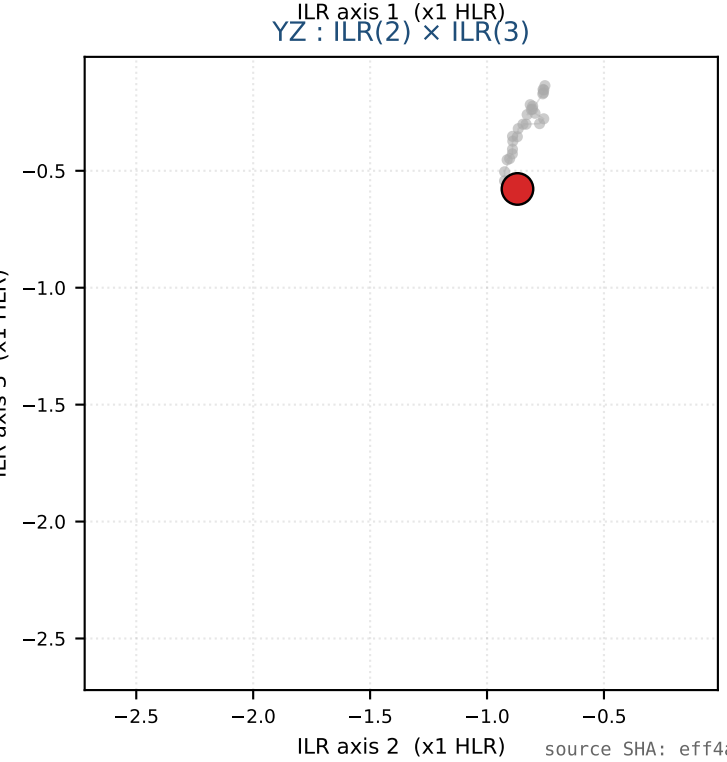
ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

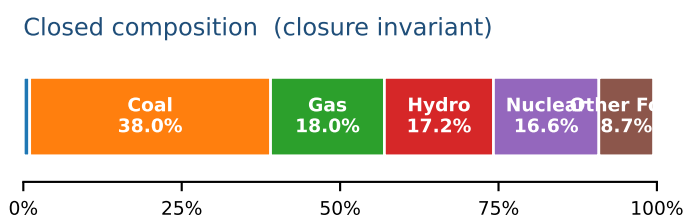




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0101	-0.713
Coal	0.3802	+2.916
Gas	0.1797	+2.167
Hydro	0.1721	+2.123
Nuclear	0.1663	+2.089
Other Fossil	0.0867	+1.437
Other Renewabl	0.0029	-1.975
Solar	0.0001	-5.731
Wind	0.0020	-2.312





closure: Σ = 1.000000 OK

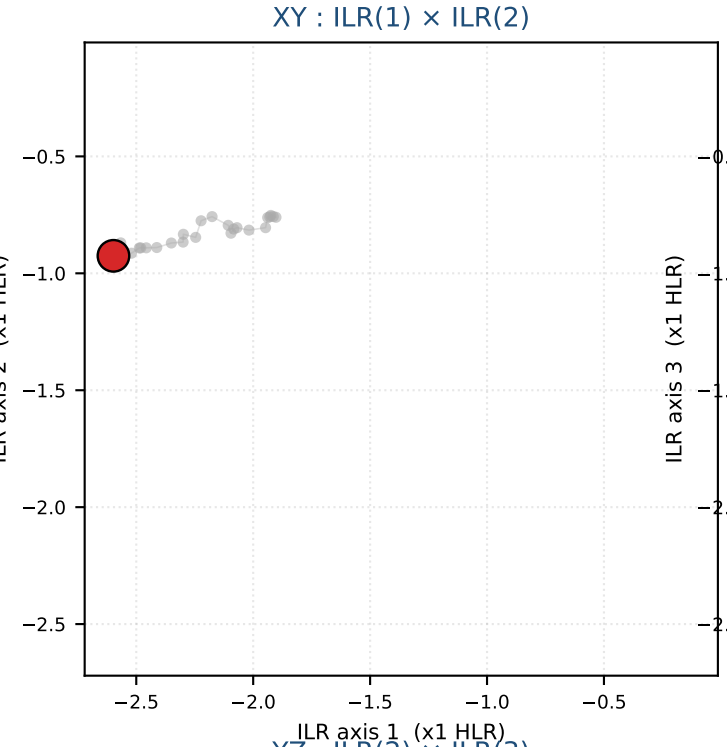
display factor: x1 (raw HLR x 1)

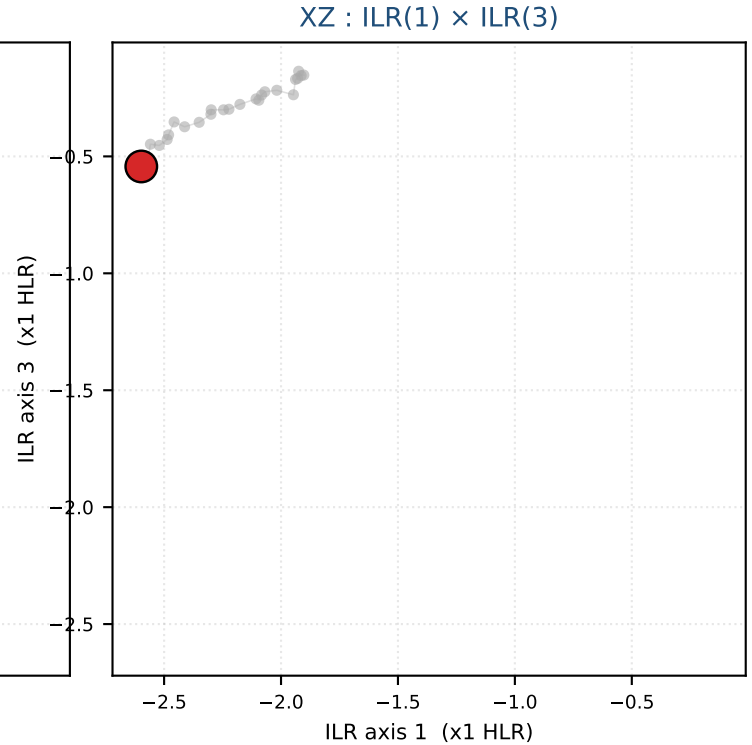
Helmert basis loadings (top 3 carriers per axis)

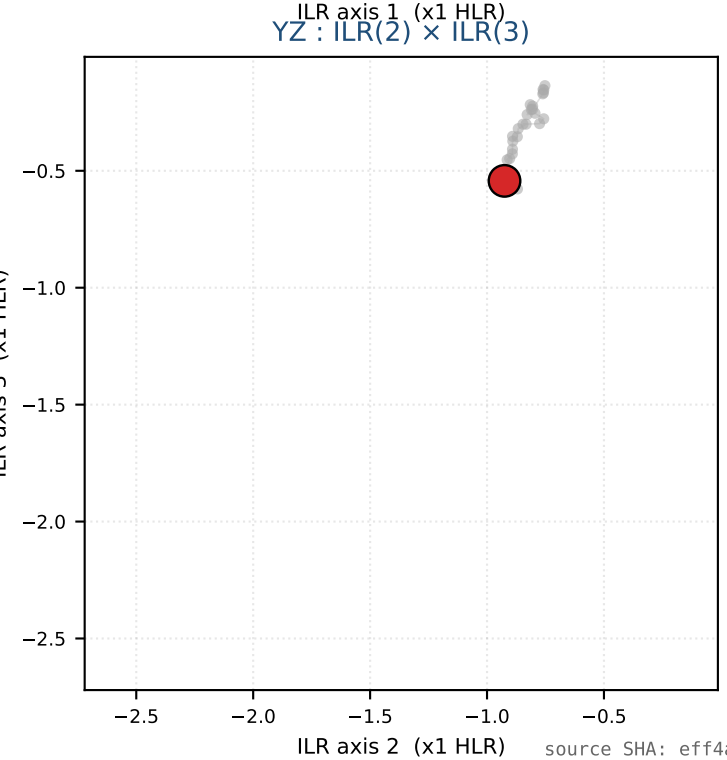
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



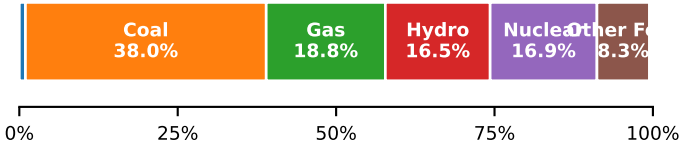




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0096	-0.801
Coal	0.3800	+2.872
Gas	0.1880	+2.169
Hydro	0.1654	+2.040
Nuclear	0.1686	+2.059
Other Fossil	0.0829	+1.350
Other Renewabl	0.0028	-2.021
Solar	0.0001	-5.501
Wind	0.0025	-2.166

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

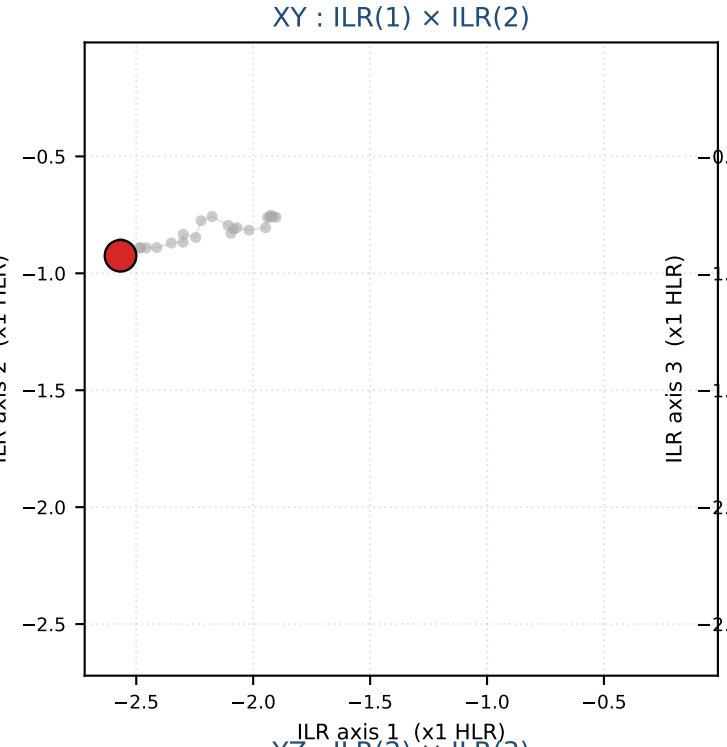
display factor: x1 (raw HLR x 1)

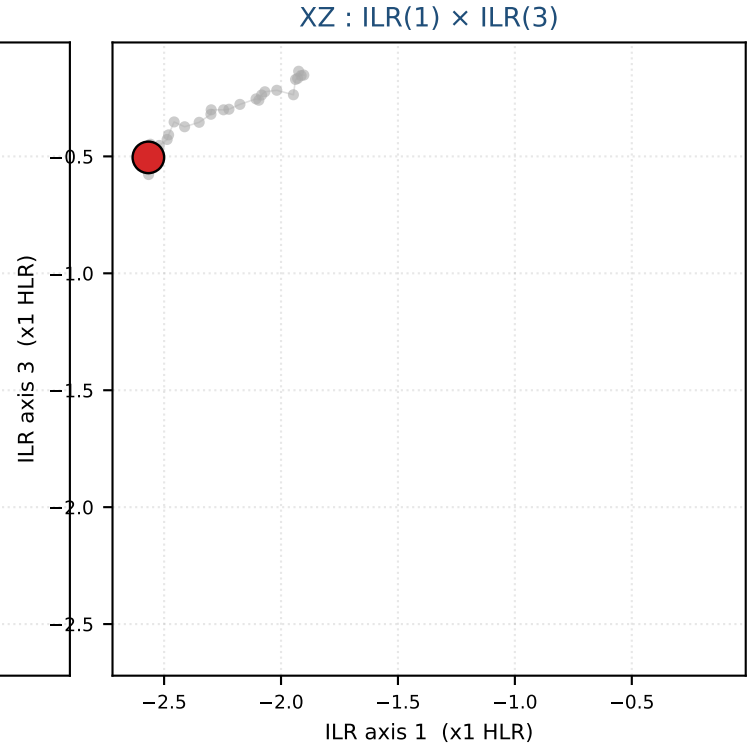
Helmert basis loadings (top 3 carriers per axis)

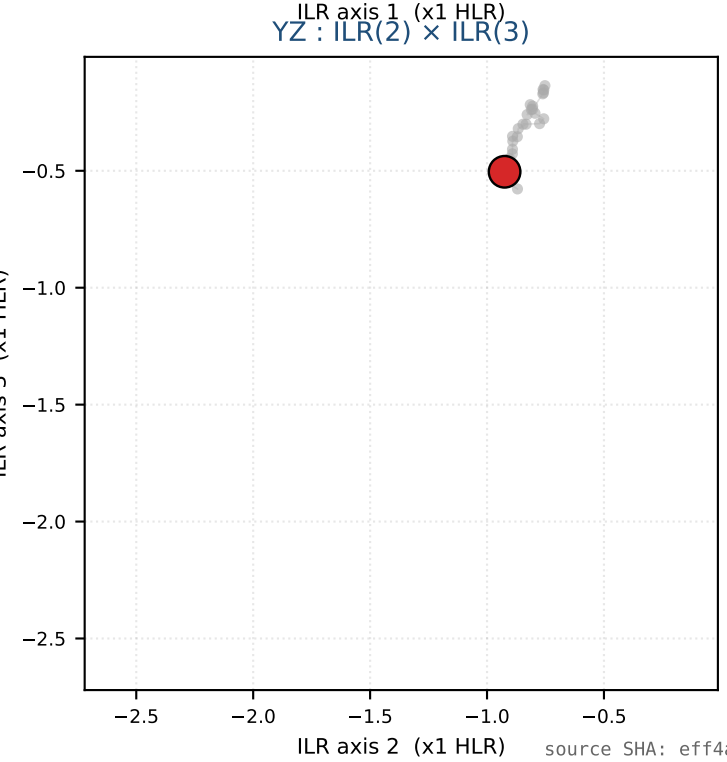
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



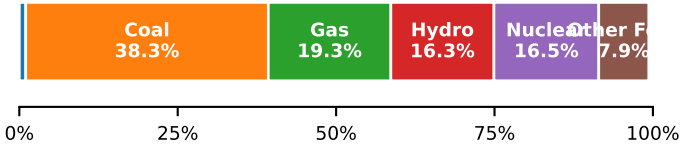




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0101	-0.805
Coal	0.3828	+2.826
Gas	0.1934	+2.143
Hydro	0.1627	+1.970
Nuclear	0.1654	+1.986
Other Fossil	0.0792	+1.251
Other Renewabl	0.0029	-2.073
Solar	0.0001	-5.355
Wind	0.0033	-1.942

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

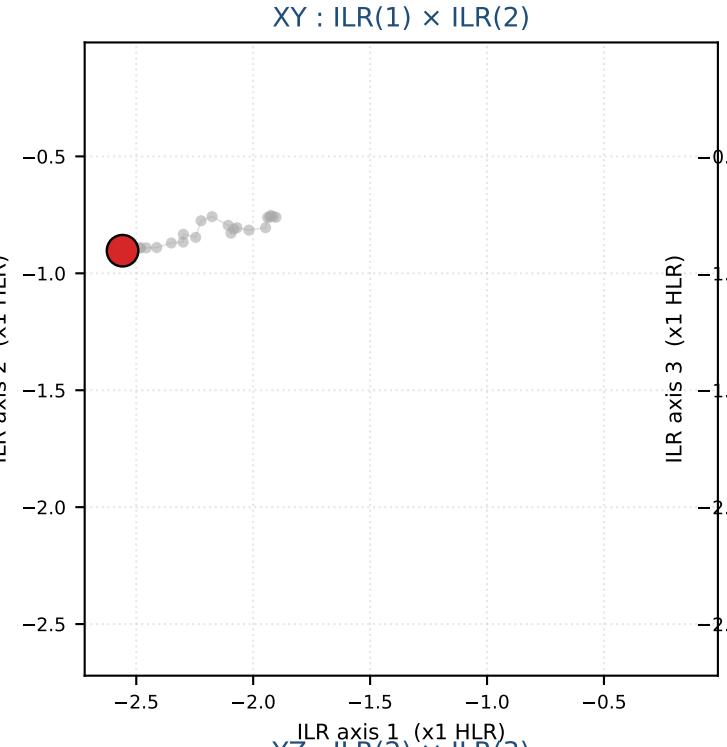
display factor: x1 (raw HLR x 1)

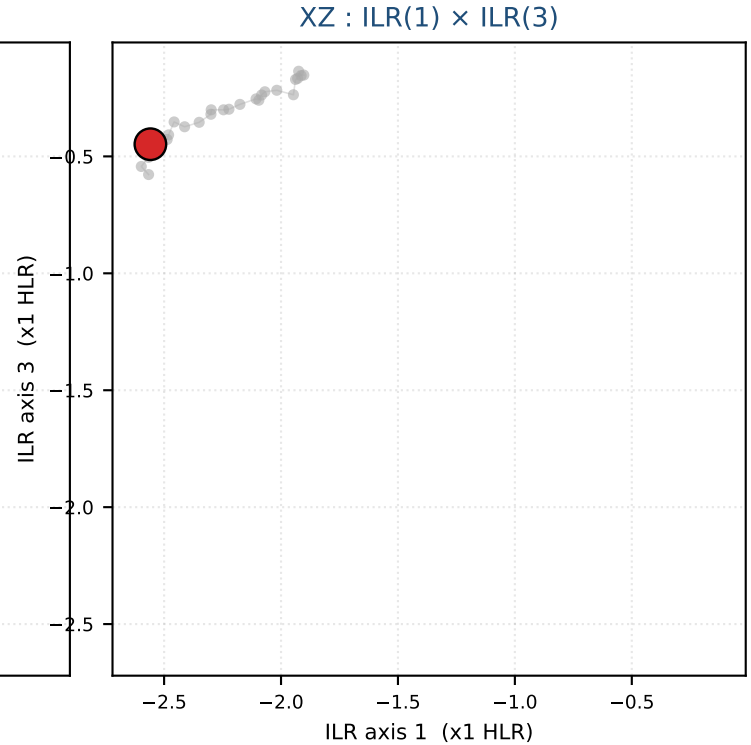
Helmert basis loadings (top 3 carriers per axis)

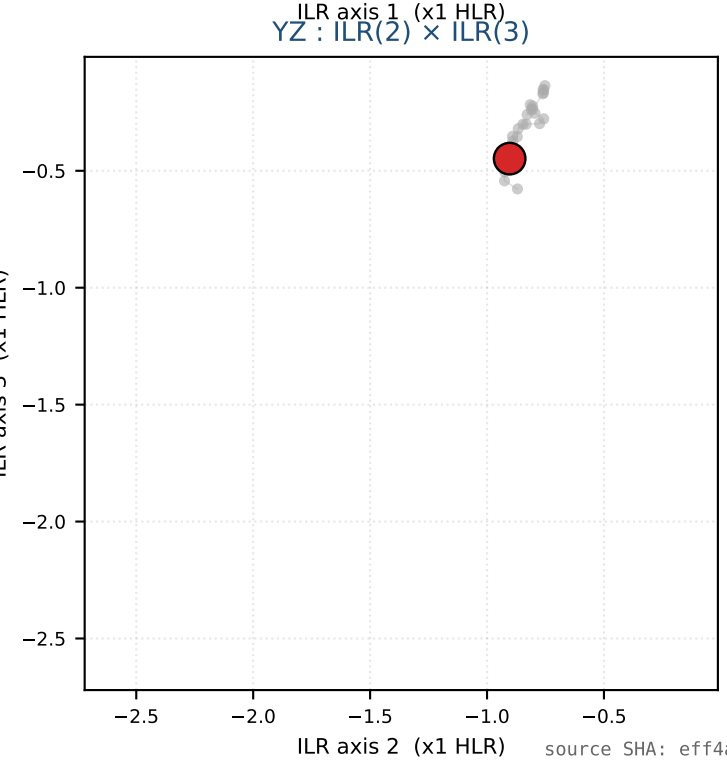
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

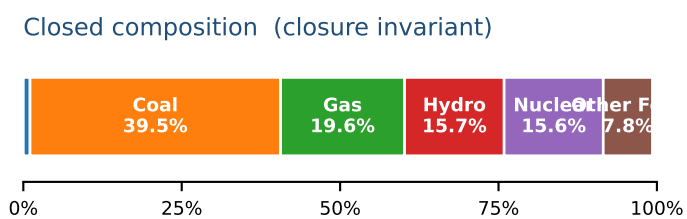






Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0106	-0.792
Coal	0.3954	+2.826
Gas	0.1958	+2.123
Hydro	0.1571	+1.903
Nuclear	0.1564	+1.899
Other Fossil	0.0780	+1.202
Other Renewabl	0.0028	-2.123
Solar	0.0001	-5.218
Wind	0.0038	-1.819



closure: Σ = 1.000000 OK

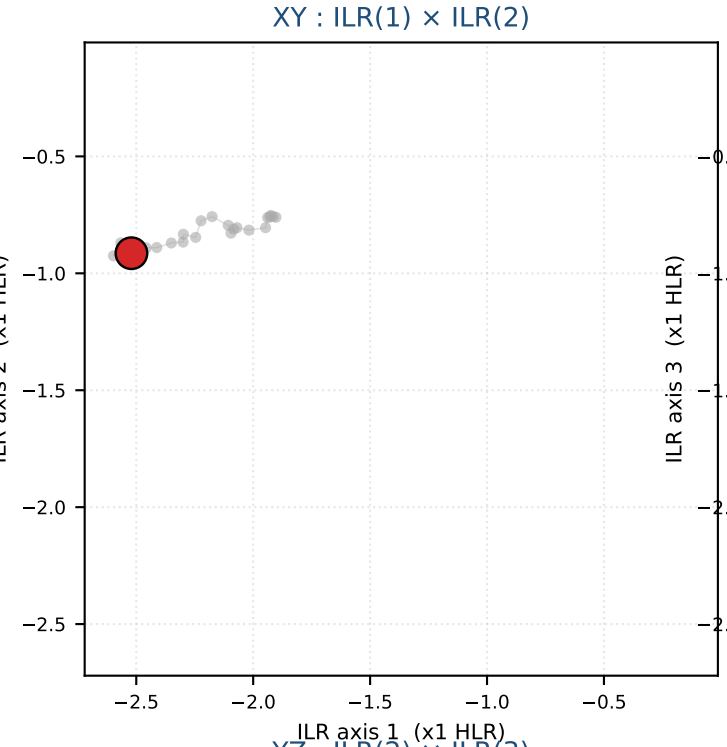
display factor: x1 (raw HLR x 1)

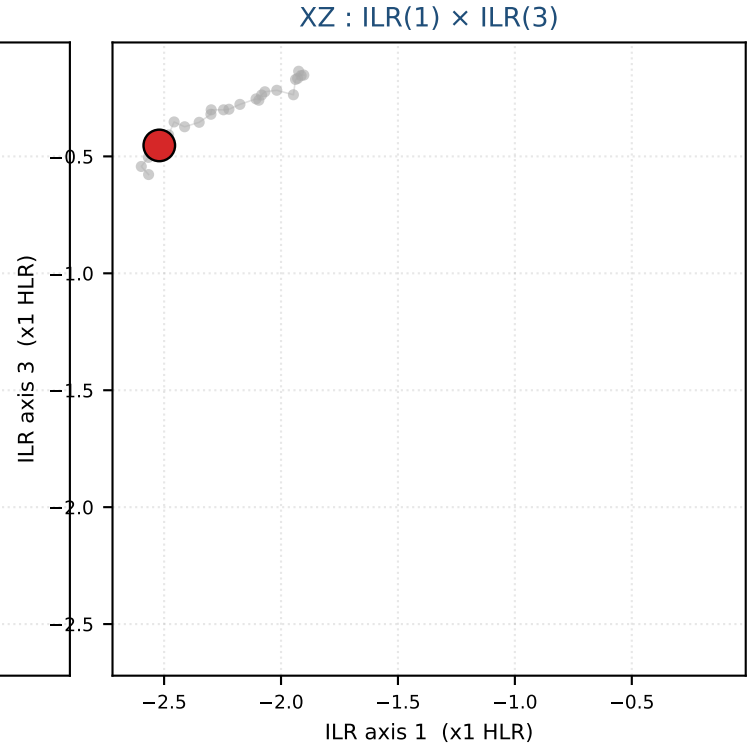
Helmert basis loadings (top 3 carriers per axis)

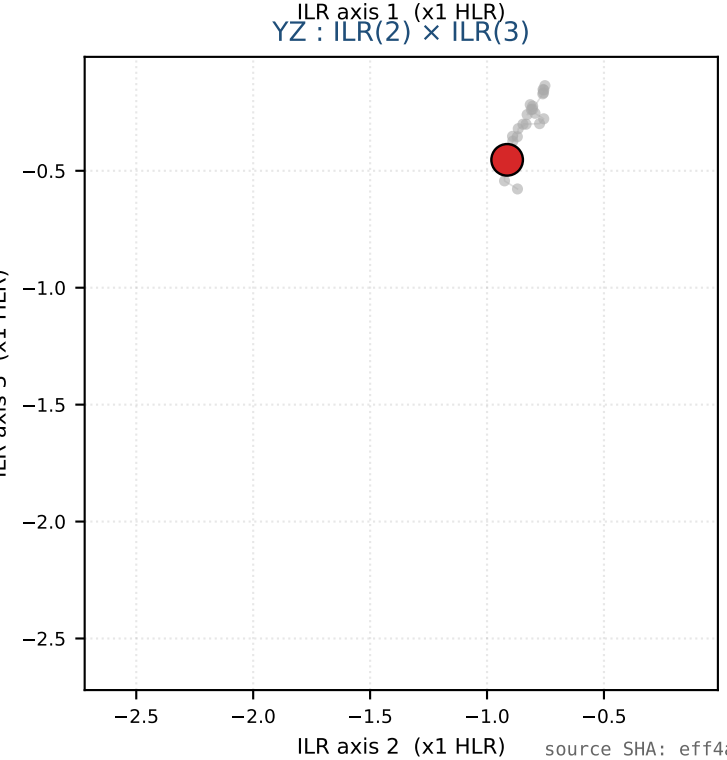
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



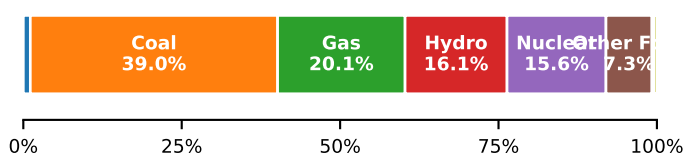




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0110	-0.805
Coal	0.3903	+2.760
Gas	0.2012	+2.097
Hydro	0.1609	+1.874
Nuclear	0.1562	+1.844
Other Fossil	0.0726	+1.077
Other Renewabl	0.0028	-2.188
Solar	0.0002	-5.042
Wind	0.0049	-1.616

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

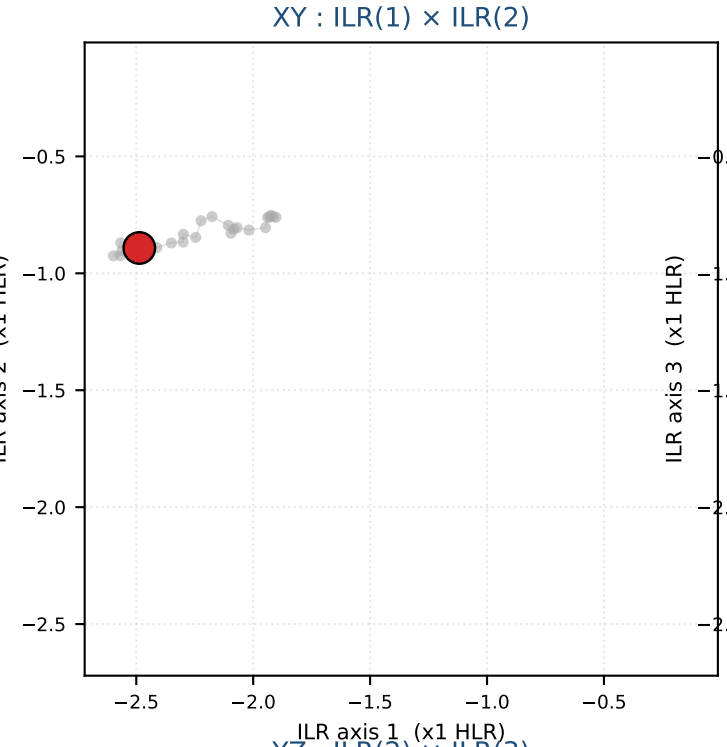
display factor: x1 (raw HLR x 1)

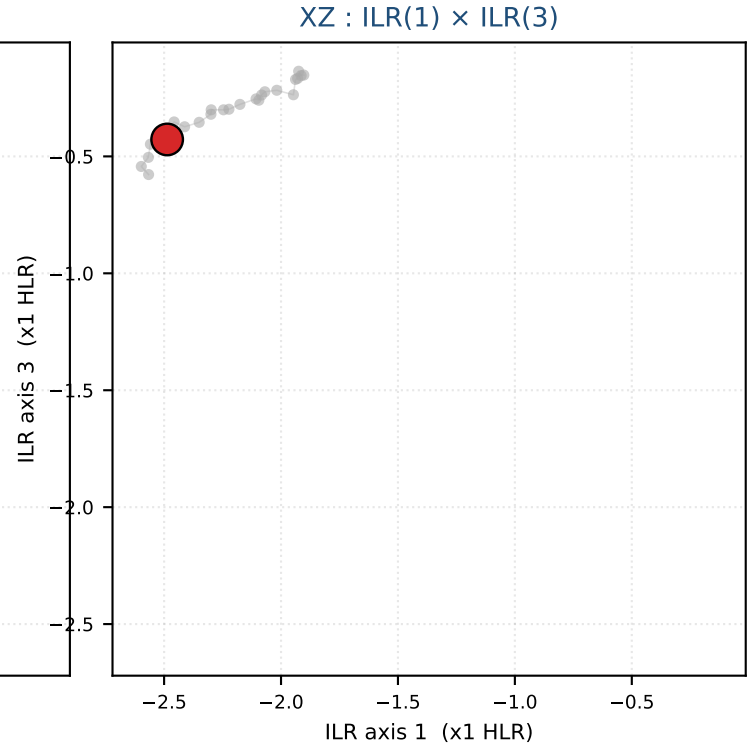
Helmert basis loadings (top 3 carriers per axis)

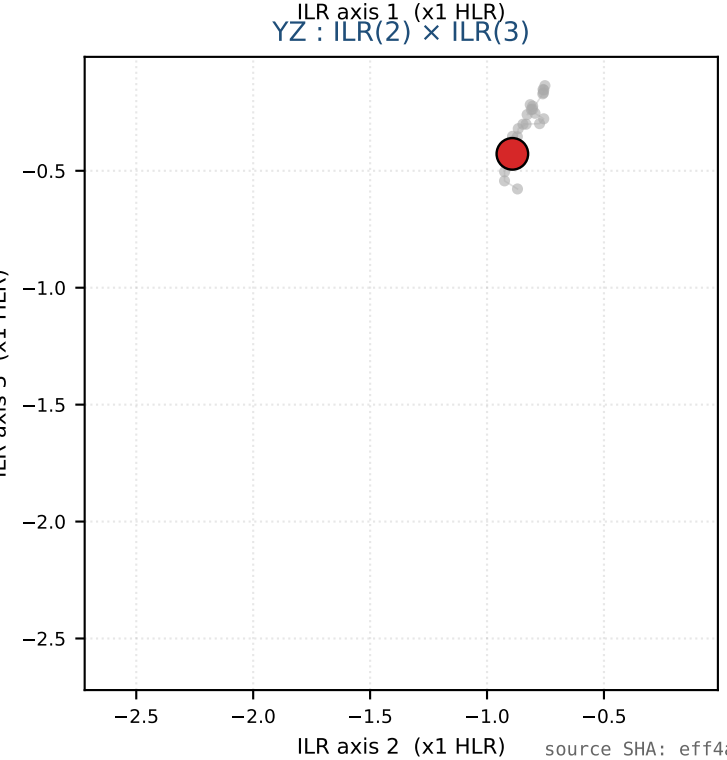
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



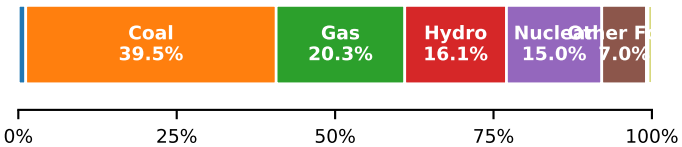




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0117	-0.795
Coal	0.3952	+2.722
Gas	0.2030	+2.056
Hydro	0.1606	+1.822
Nuclear	0.1504	+1.756
Other Fossil	0.0705	+0.998
Other Renewabl	0.0027	-2.271
Solar	0.0002	-4.781
Wind	0.0058	-1.507

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

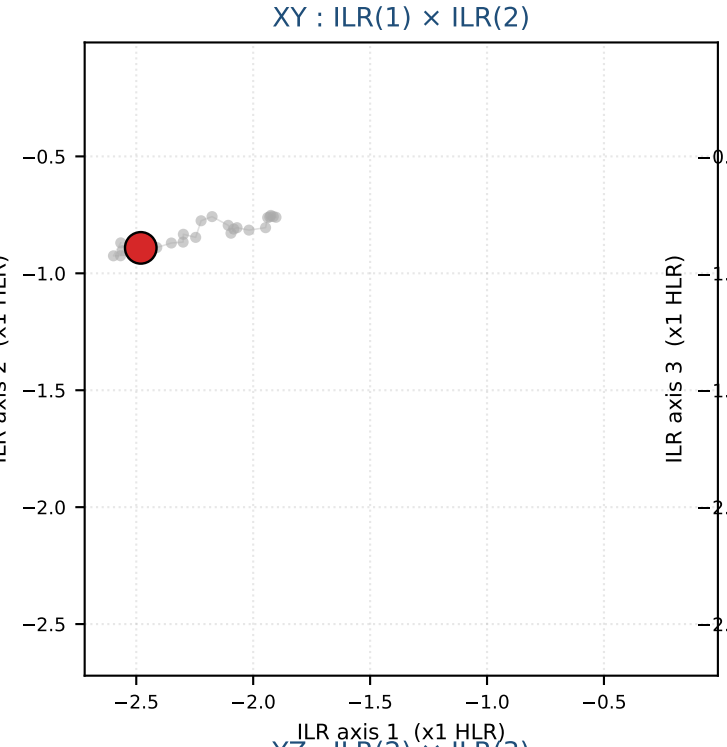
display factor: x1 (raw HLR x 1)

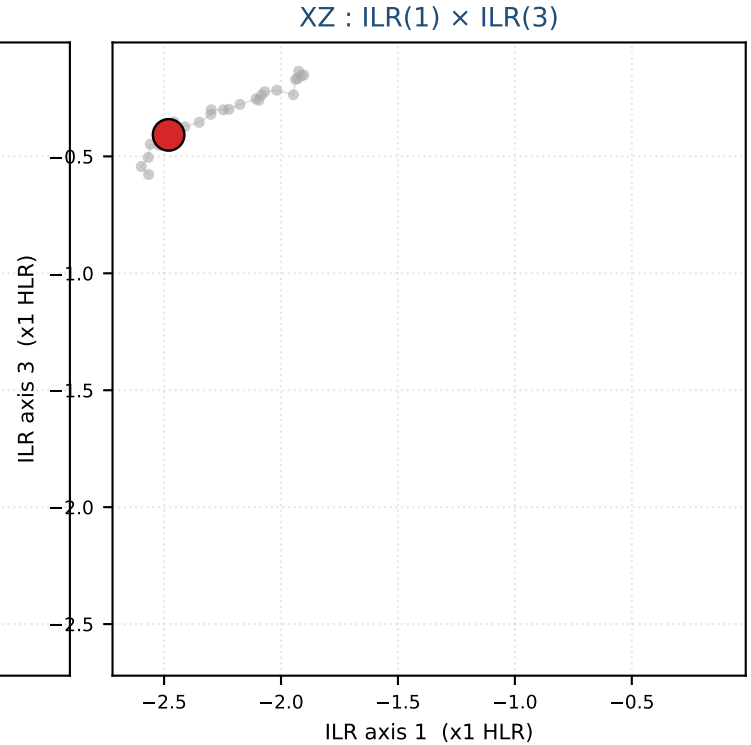
Helmert basis loadings (top 3 carriers per axis)

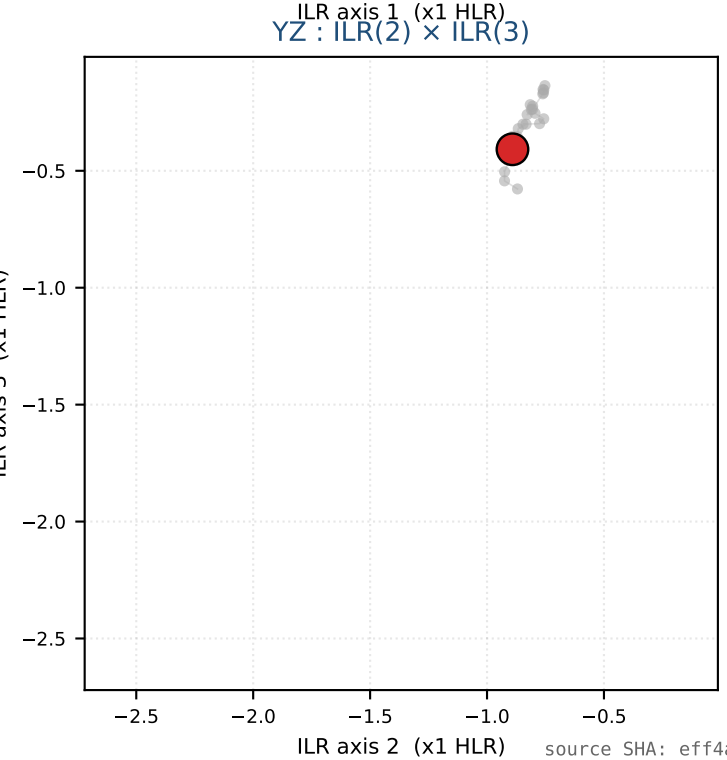
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

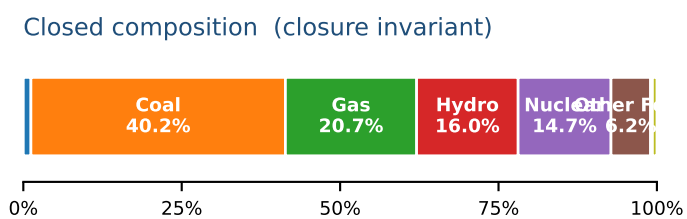






Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0120	-0.814
Coal	0.4017	+2.694
Gas	0.2070	+2.031
Hydro	0.1602	+1.775
Nuclear	0.1466	+1.686
Other Fossil	0.0624	+0.832
Other Renewabl	0.0027	-2.310
Solar	0.0003	-4.547
Wind	0.0071	-1.346



closure: Σ = 1.000000 OK

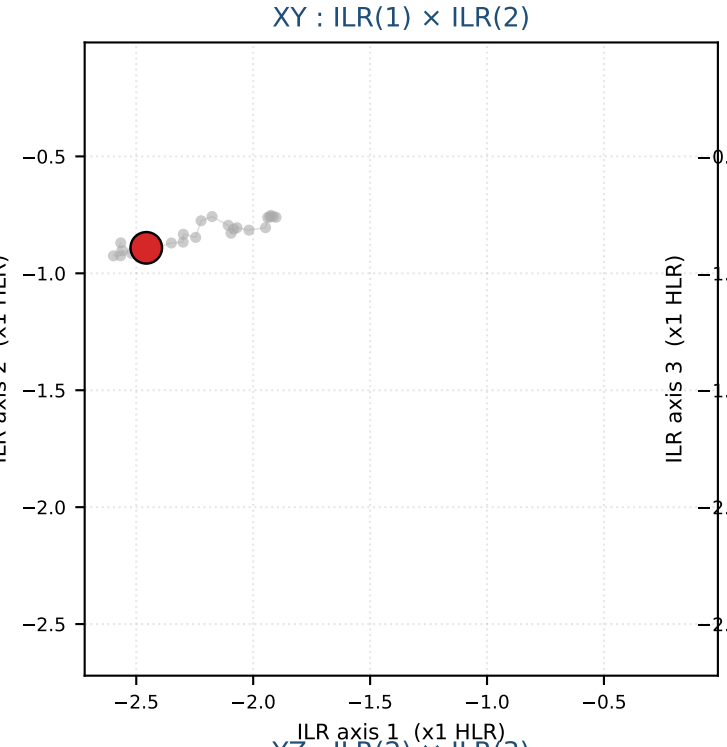
display factor: x1 (raw HLR x 1)

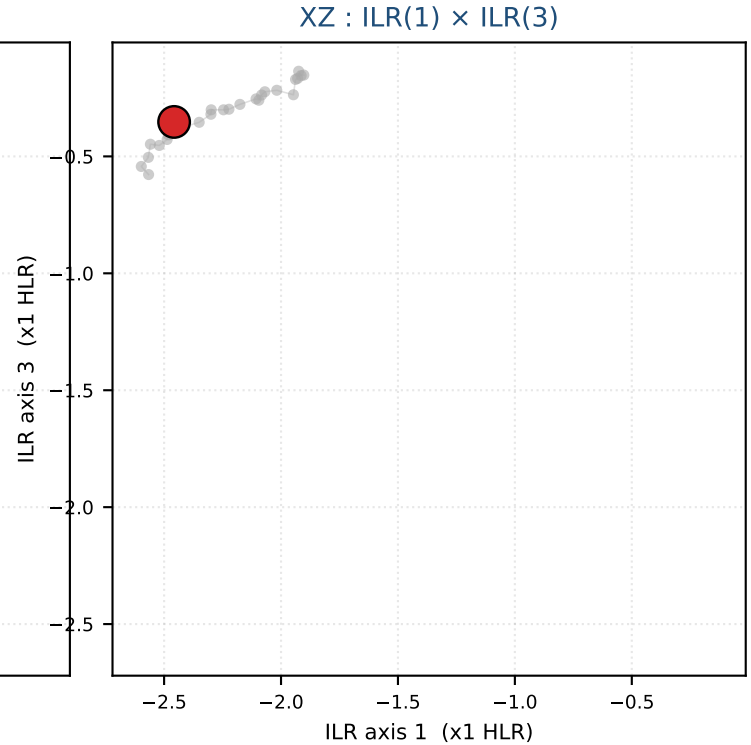
Helmert basis loadings (top 3 carriers per axis)

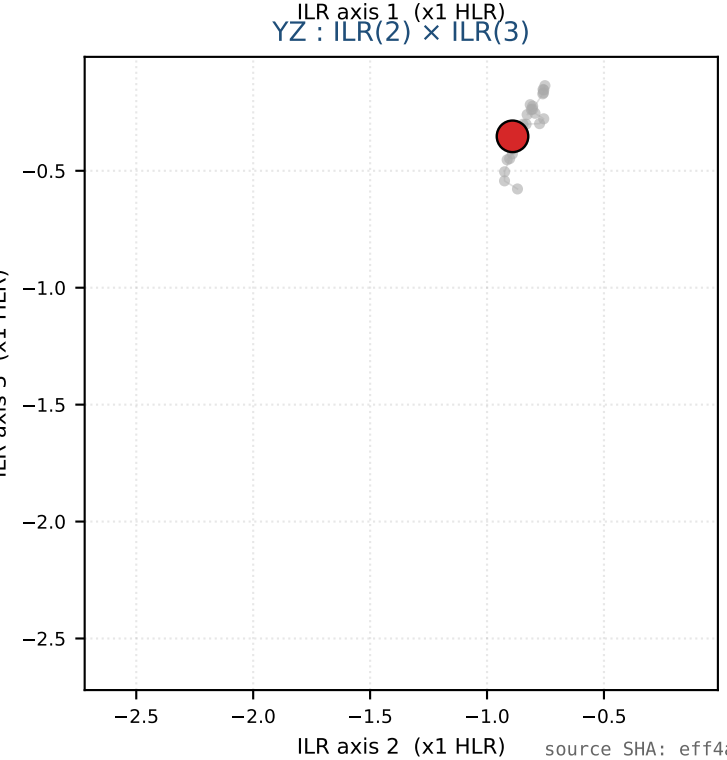
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



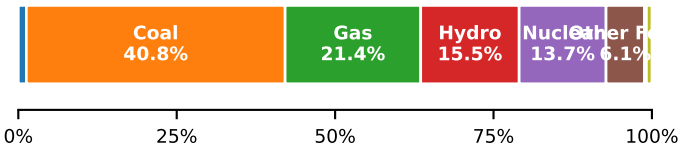




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0126	-0.812
Coal	0.4084	+2.663
Gas	0.2140	+2.017
Hydro	0.1554	+1.697
Nuclear	0.1371	+1.572
Other Fossil	0.0607	+0.757
Other Renewabl	0.0027	-2.363
Solar	0.0004	-4.344
Wind	0.0087	-1.188

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

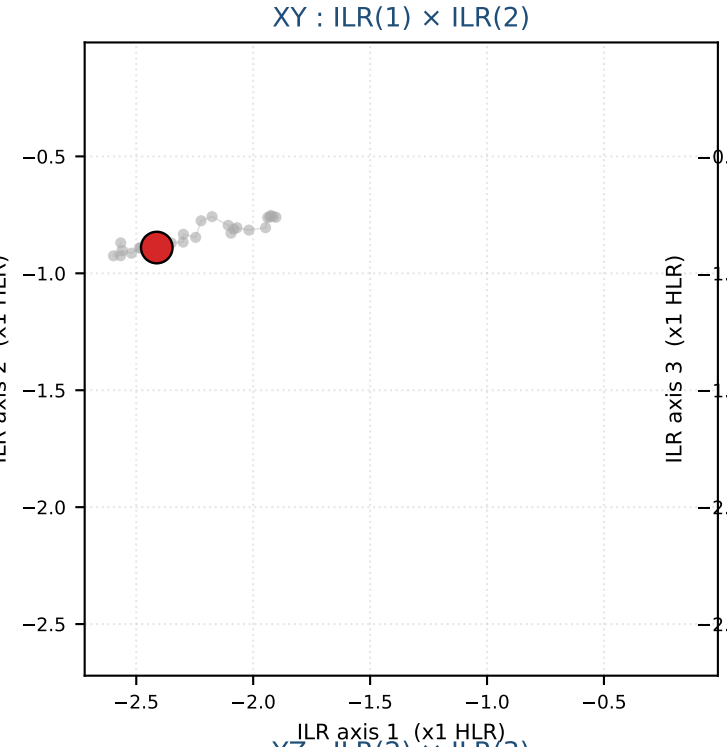
display factor: x1 (raw HLR x 1)

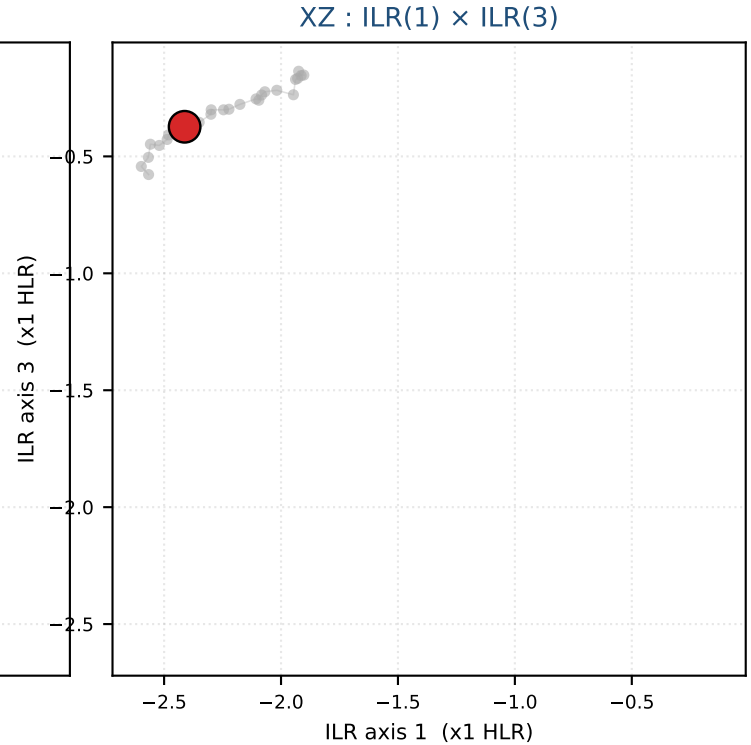
Helmert basis loadings (top 3 carriers per axis)

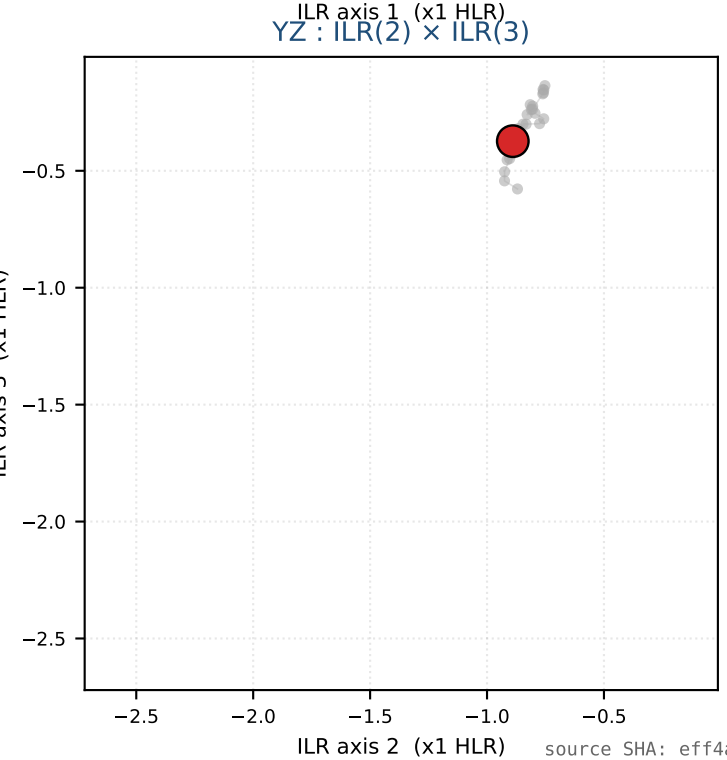
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



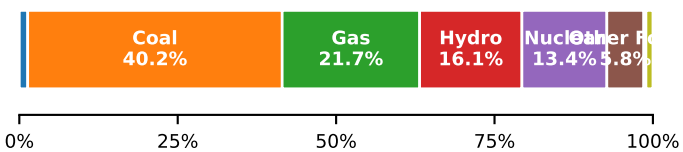




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0133	-0.850
Coal	0.4016	+2.561
Gas	0.2169	+1.945
Hydro	0.1614	+1.650
Nuclear	0.1340	+1.464
Other Fossil	0.0583	+0.631
Other Renewabl	0.0028	-2.399
Solar	0.0006	-3.963
Wind	0.0110	-1.038

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

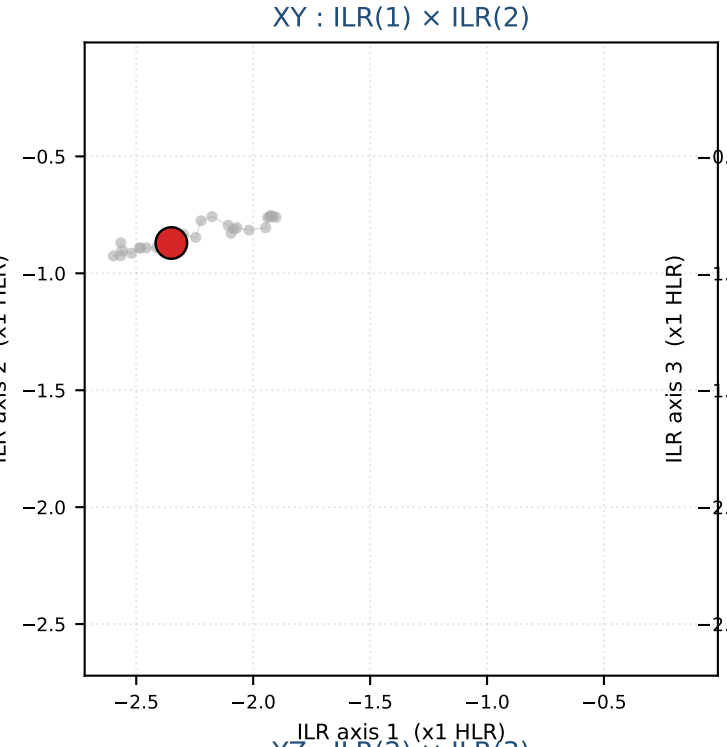
display factor: x1 (raw HLR x 1)

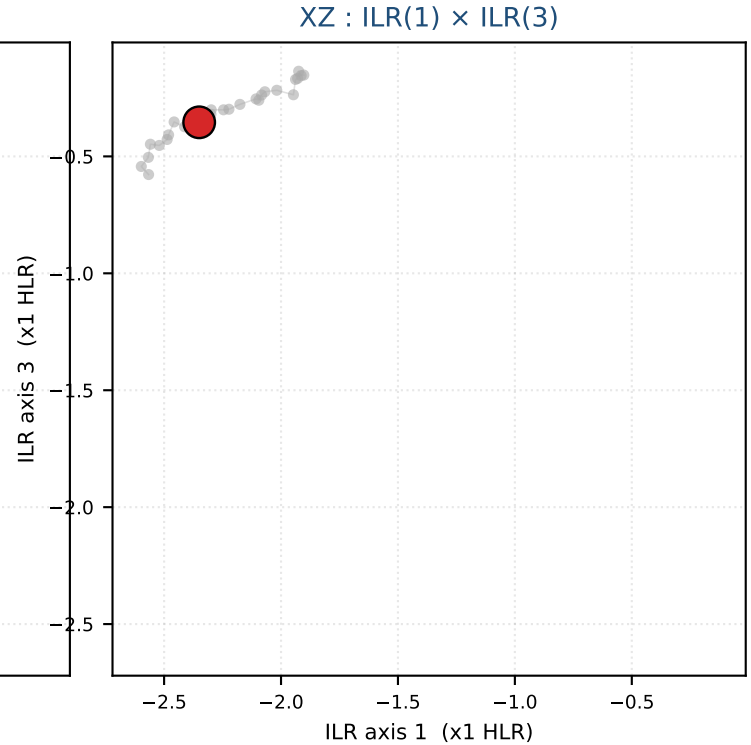
Helmert basis loadings (top 3 carriers per axis)

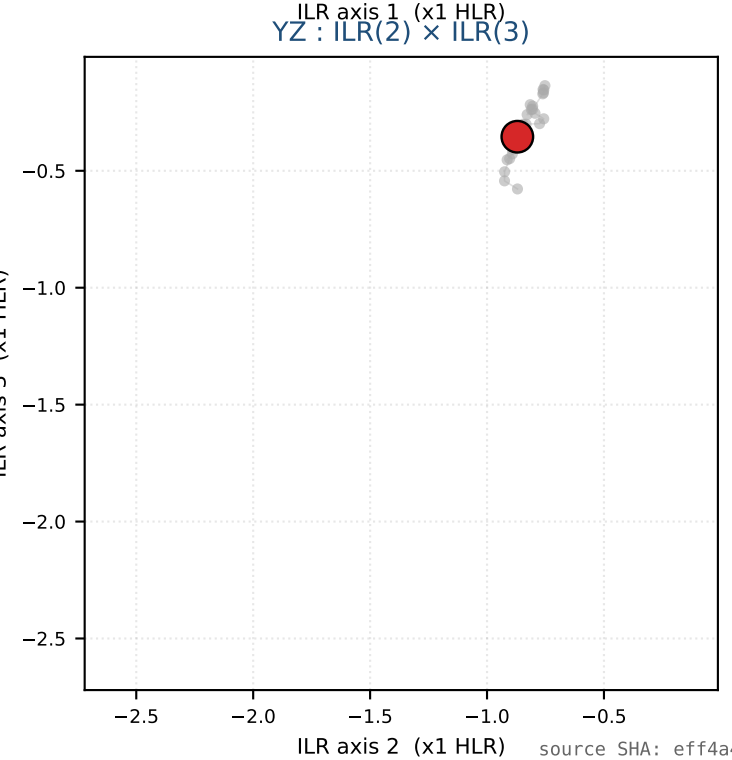
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



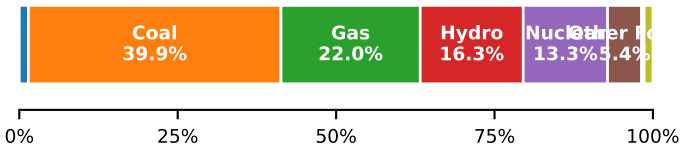




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0144	-0.859
Coal	0.3987	+2.464
Gas	0.2197	+1.869
Hydro	0.1625	+1.567
Nuclear	0.1332	+1.368
Other Fossil	0.0537	+0.460
Other Renewabl	0.0029	-2.445
Solar	0.0010	-3.530
Wind	0.0139	-0.895

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

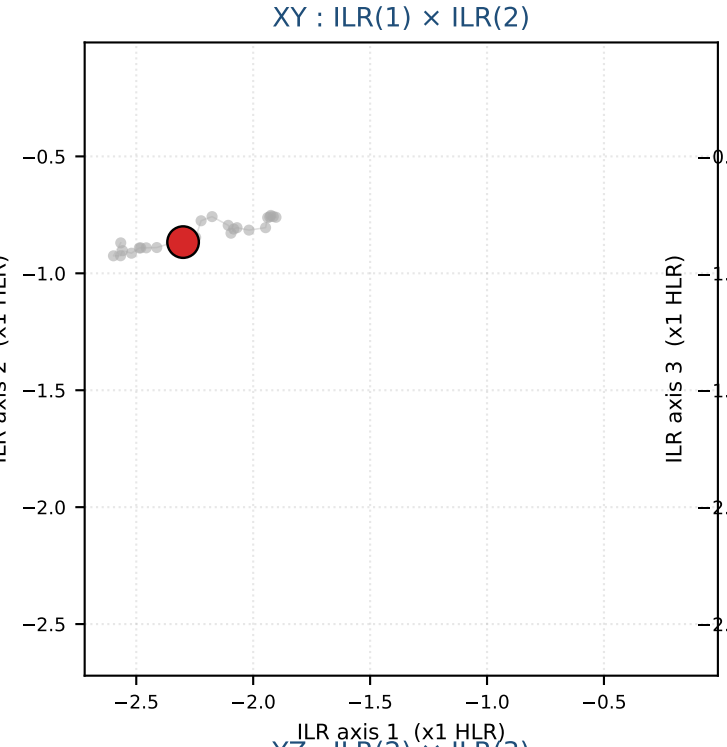
display factor: x1 (raw HLR x 1)

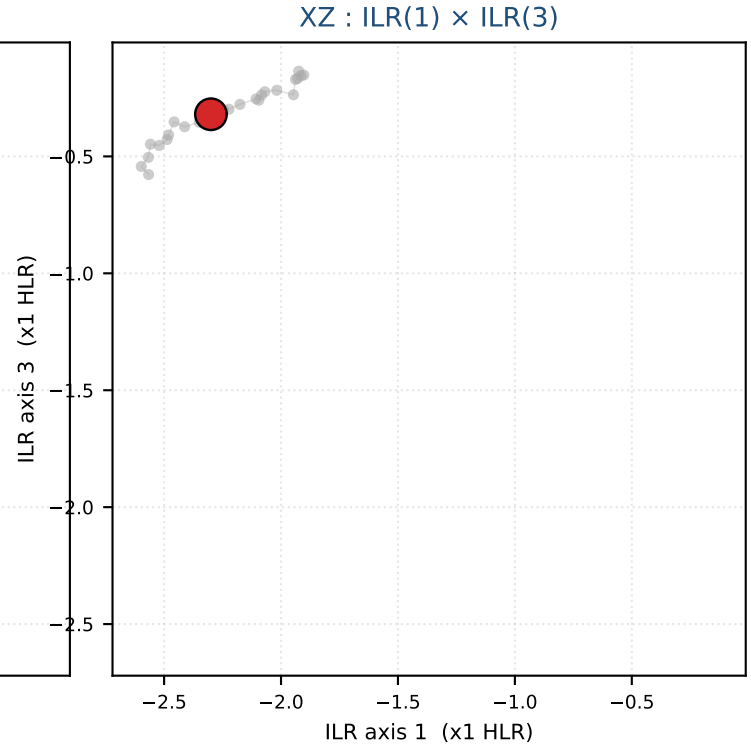
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=9 T=26 | level 1 first-principles

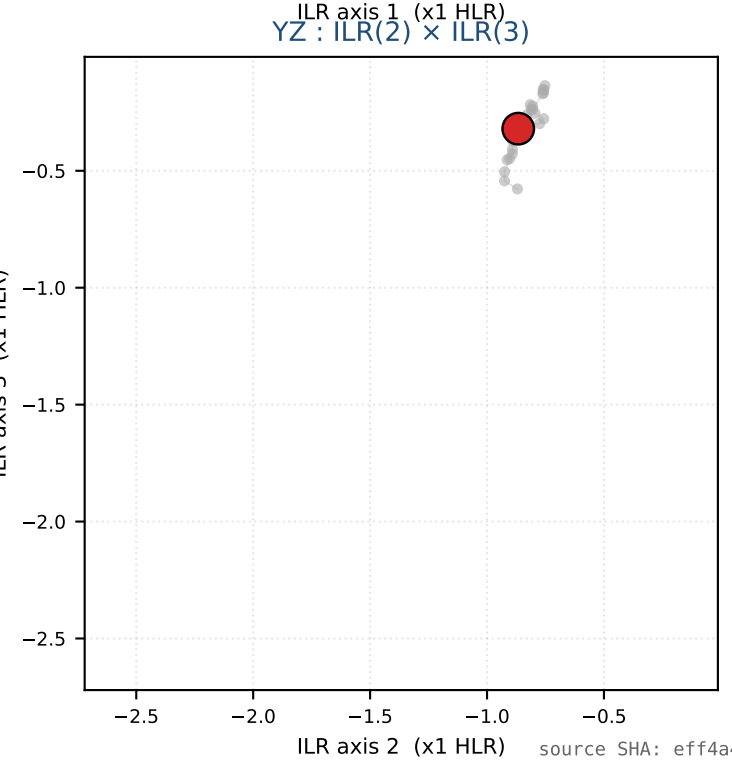
ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

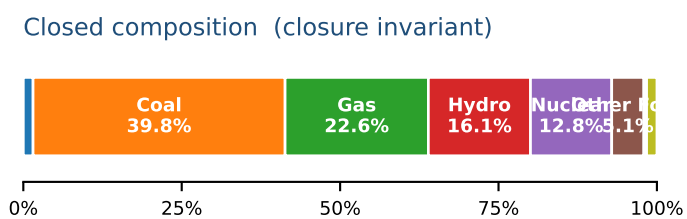




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0154	-0.849
Coal	0.3978	+2.404
Gas	0.2261	+1.839
Hydro	0.1612	+1.501
Nuclear	0.1282	+1.271
Other Fossil	0.0507	+0.344
Other Renewabl	0.0028	-2.550
Solar	0.0015	-3.167
Wind	0.0163	-0.793





closure: Σ = 1.000000 OK

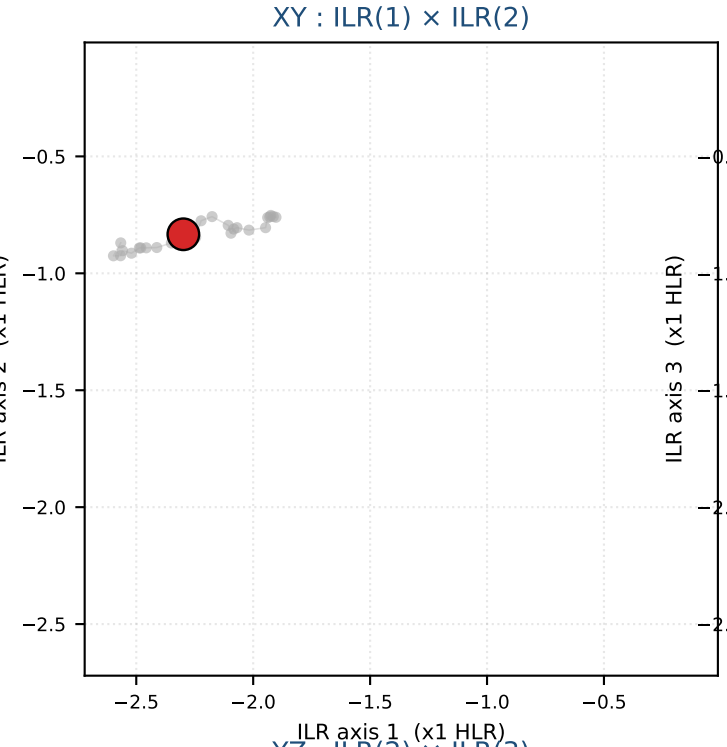
display factor: x1 (raw HLR x 1)

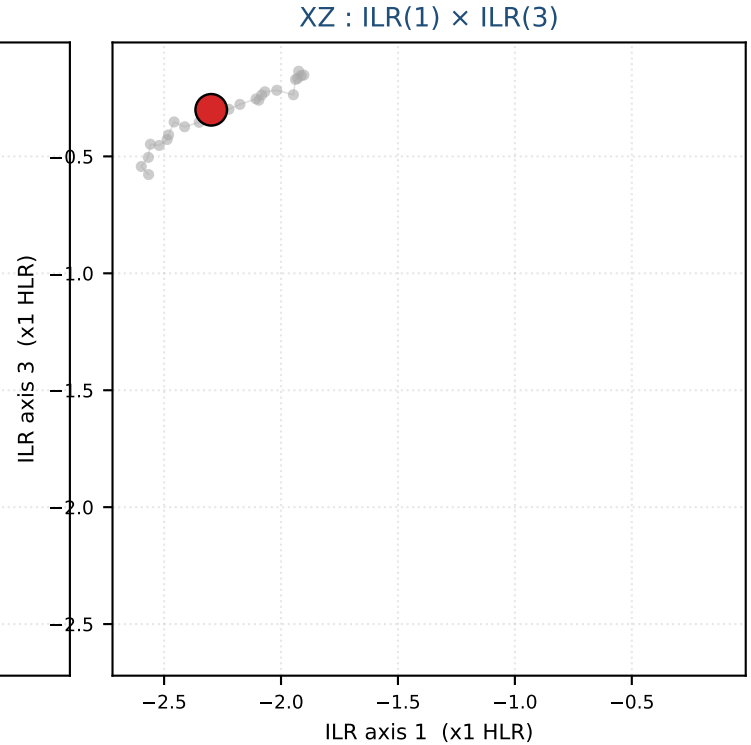
Helmert basis loadings (top 3 carriers per axis)

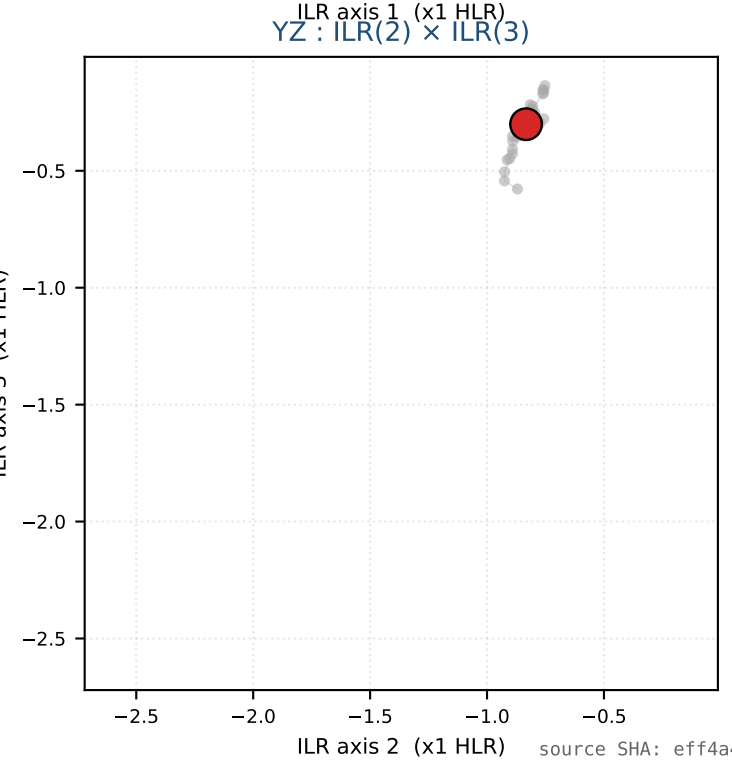
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



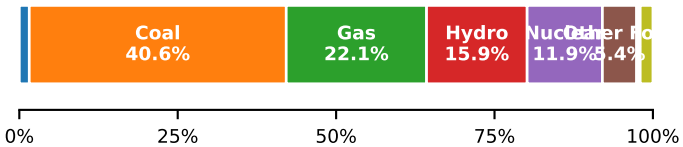




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0157	-0.921
Coal	0.4056	+2.330
Gas	0.2215	+1.725
Hydro	0.1588	+1.392
Nuclear	0.1188	+1.102
Other Fossil	0.0540	+0.314
Other Renewabl	0.0028	-2.653
Solar	0.0029	-2.613
Wind	0.0200	-0.678

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

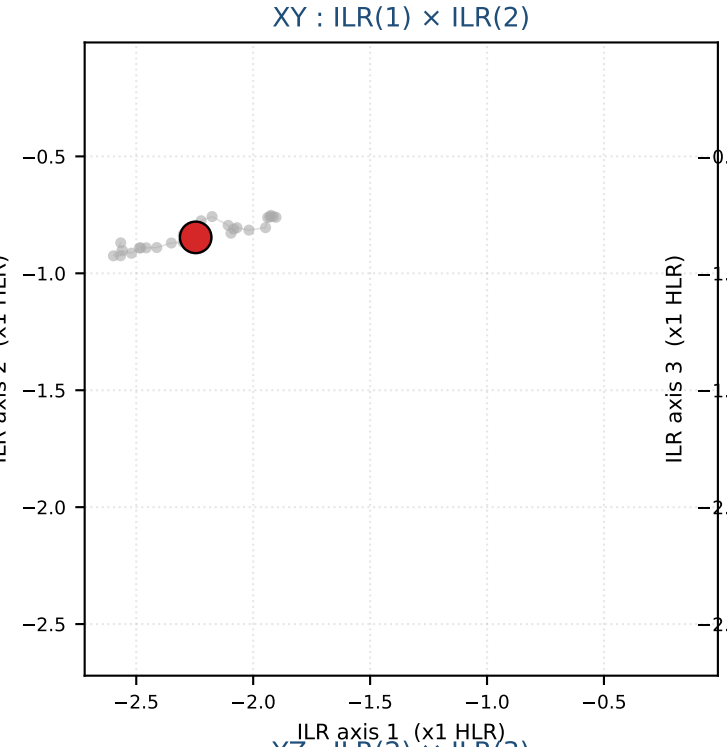
display factor: x1 (raw HLR x 1)

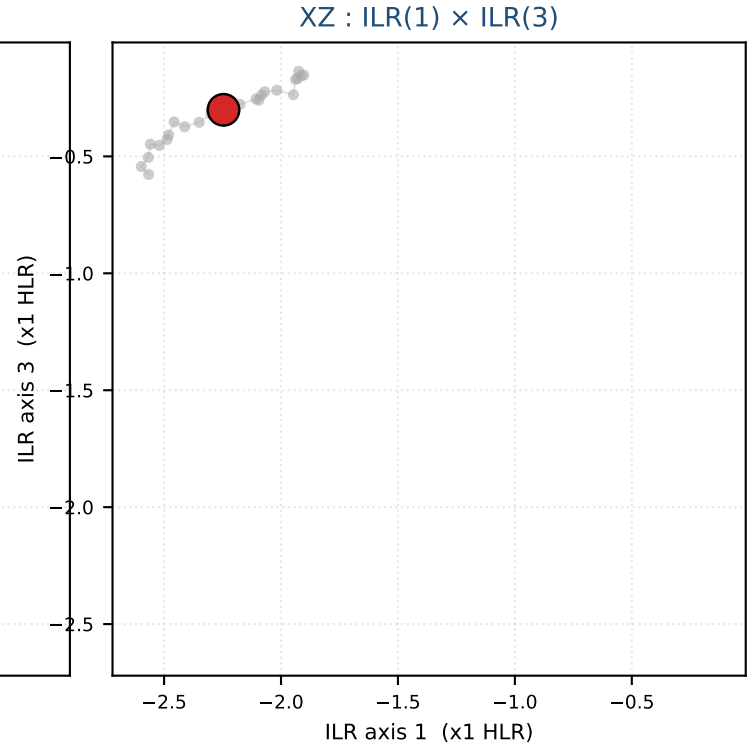
Helmert basis loadings (top 3 carriers per axis)

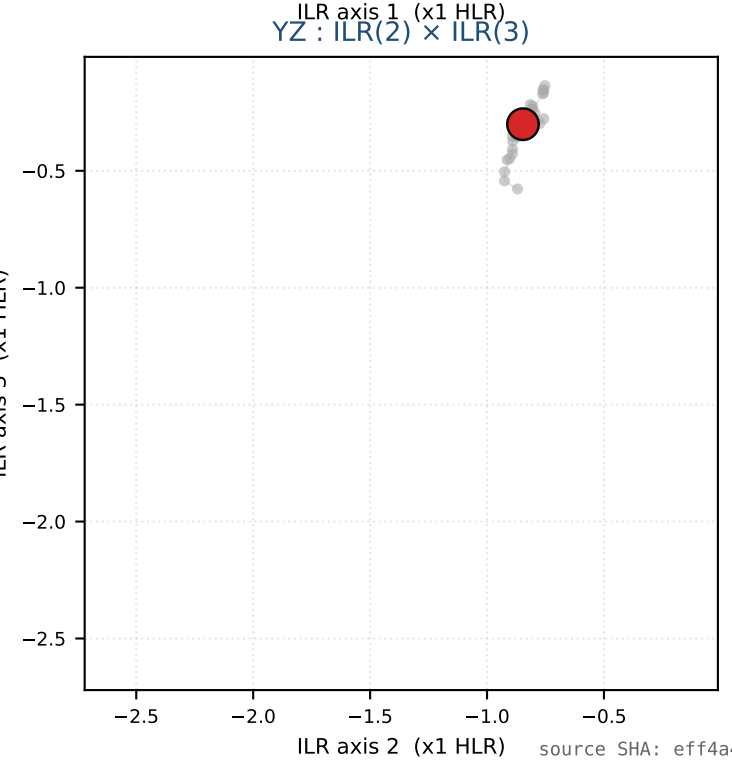
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



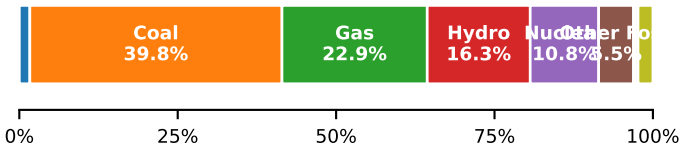




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0166	-0.929
Coal	0.3980	+2.247
Gas	0.2292	+1.695
Hydro	0.1626	+1.352
Nuclear	0.1080	+0.943
Other Fossil	0.0550	+0.268
Other Renewabl	0.0028	-2.713
Solar	0.0043	-2.280
Wind	0.0235	-0.582

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

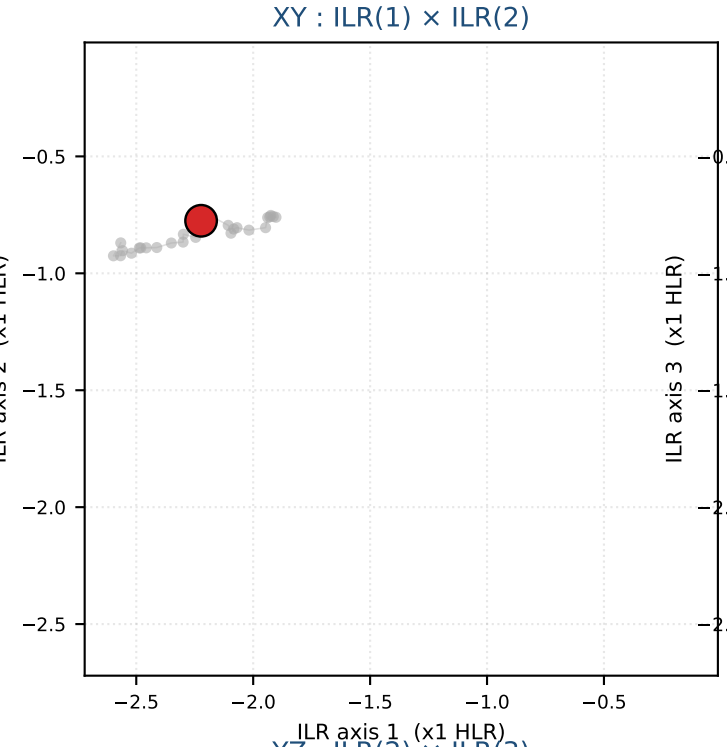
display factor: x1 (raw HLR x 1)

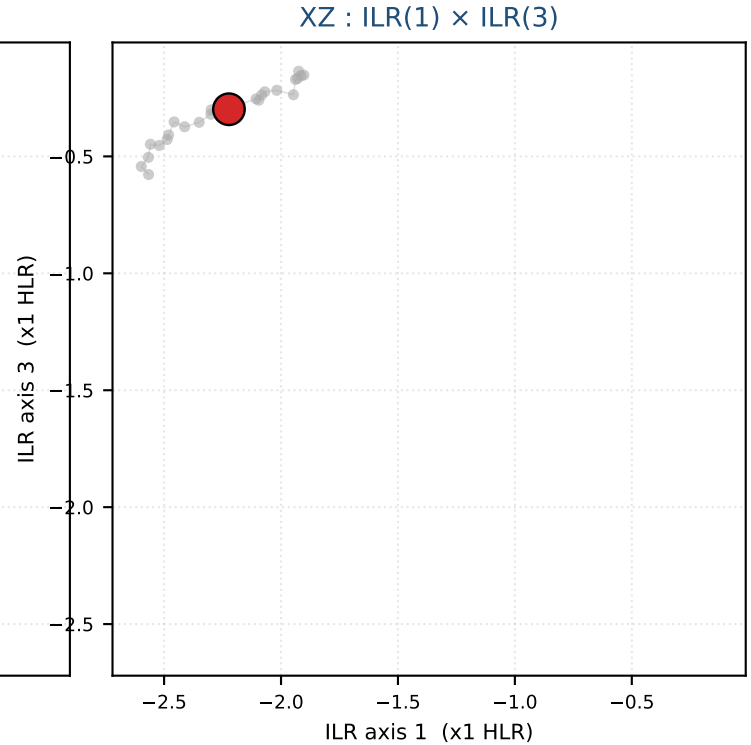
Helmert basis loadings (top 3 carriers per axis)

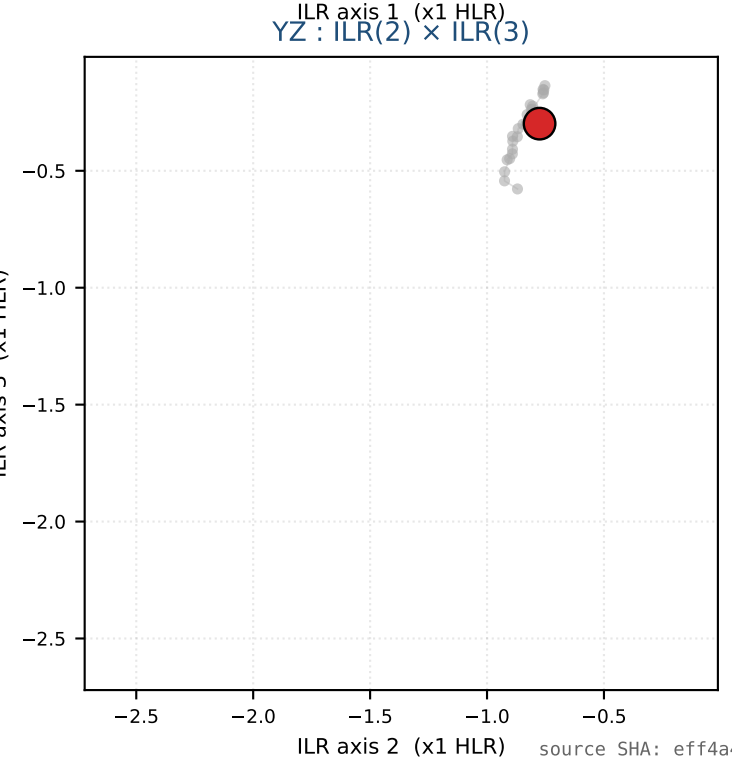
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

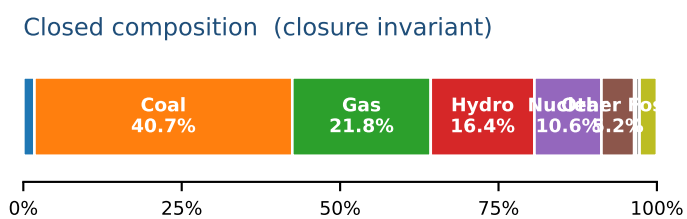






Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0176	-0.916
Coal	0.4070	+2.227
Gas	0.2185	+1.605
Hydro	0.1637	+1.316
Nuclear	0.1057	+0.879
Other Fossil	0.0517	+0.164
Other Renewabl	0.0028	-2.760
Solar	0.0057	-2.043
Wind	0.0274	-0.472



closure: Σ = 1.000000 OK

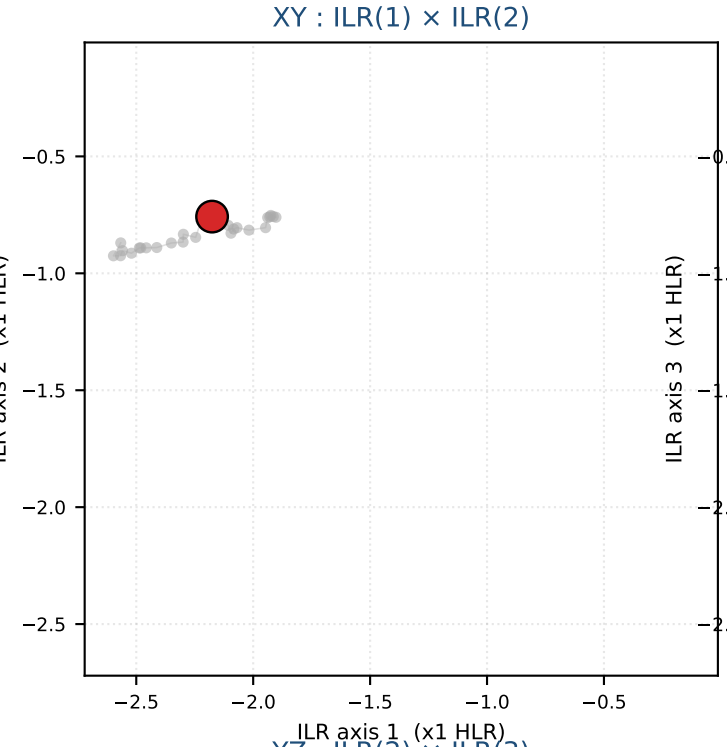
display factor: x1 (raw HLR x 1)

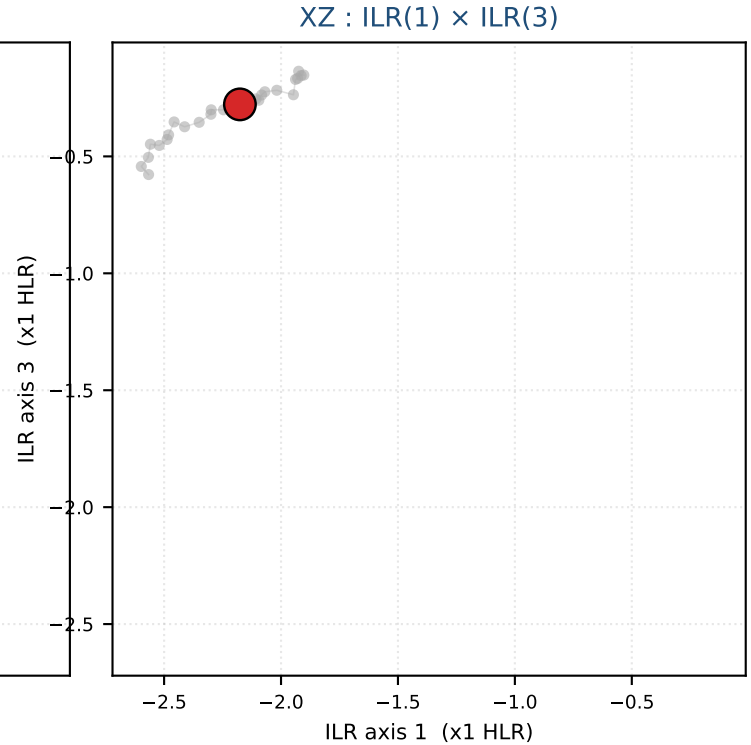
Helmert basis loadings (top 3 carriers per axis)

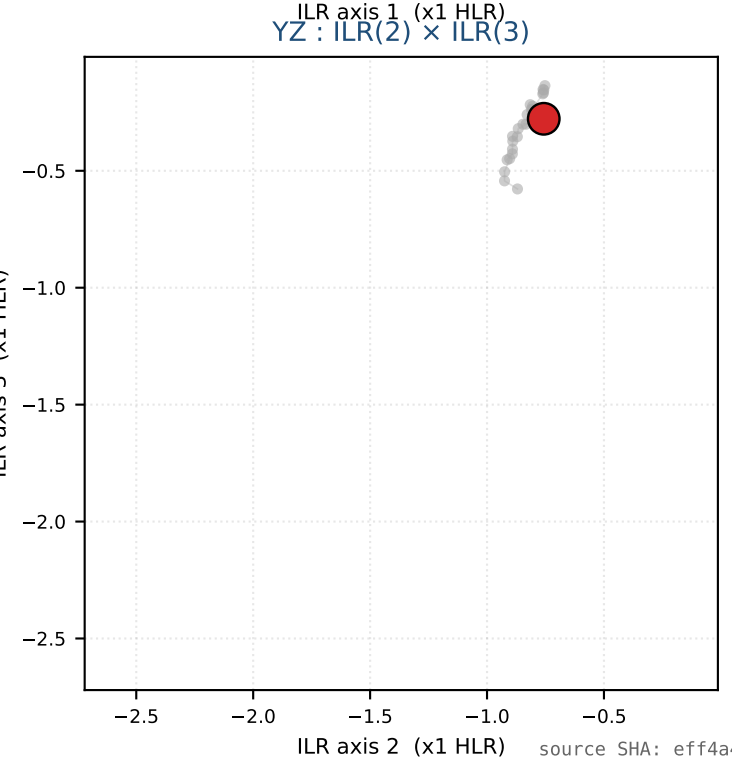
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



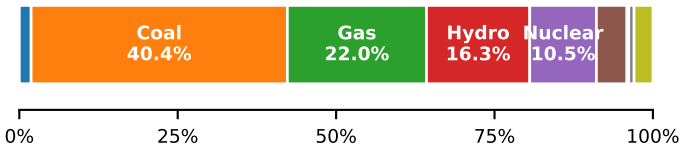




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0186	-0.911
Coal	0.4044	+2.166
Gas	0.2196	+1.555
Hydro	0.1630	+1.257
Nuclear	0.1052	+0.819
Other Fossil	0.0482	+0.038
Other Renewabl	0.0029	-2.762
Solar	0.0083	-1.717
Wind	0.0297	-0.444

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

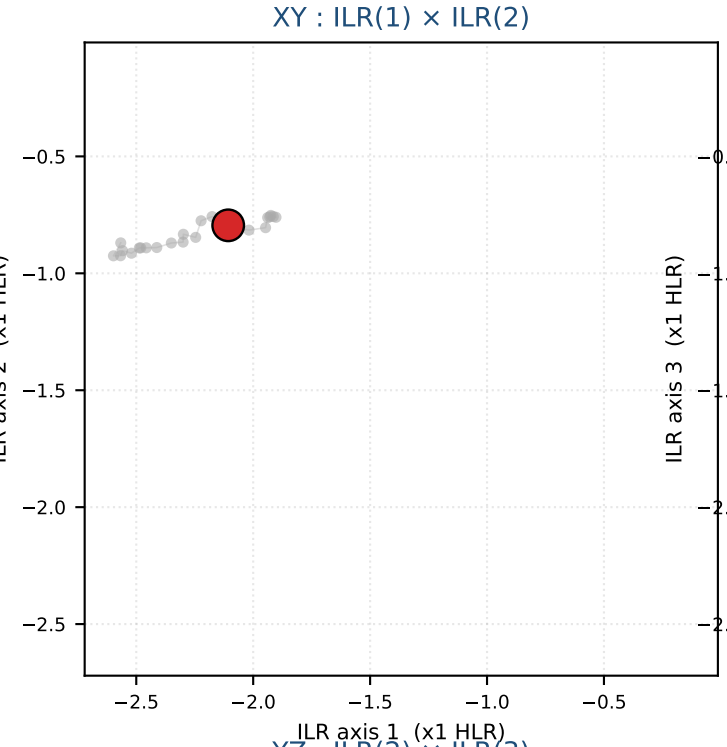
display factor: x1 (raw HLR x 1)

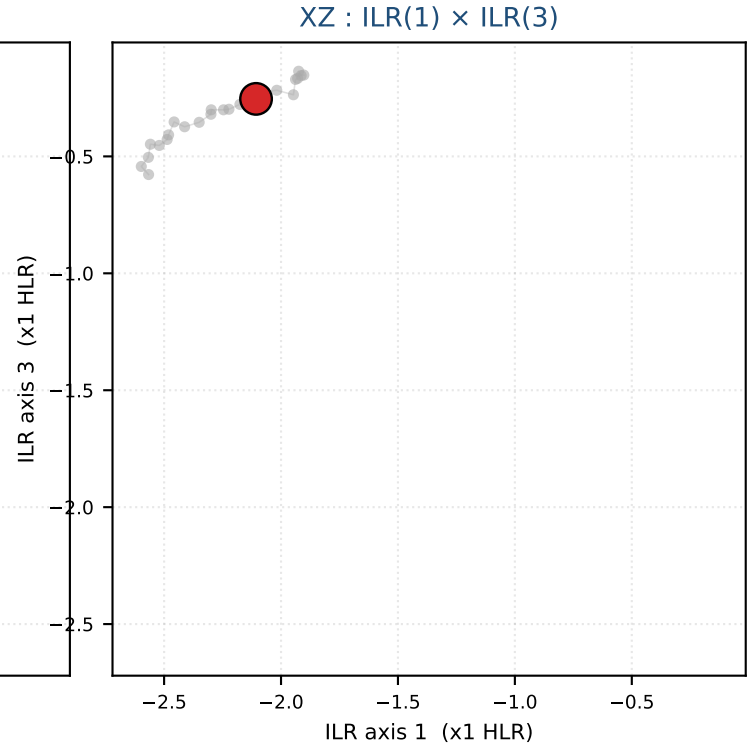
Helmert basis loadings (top 3 carriers per axis)

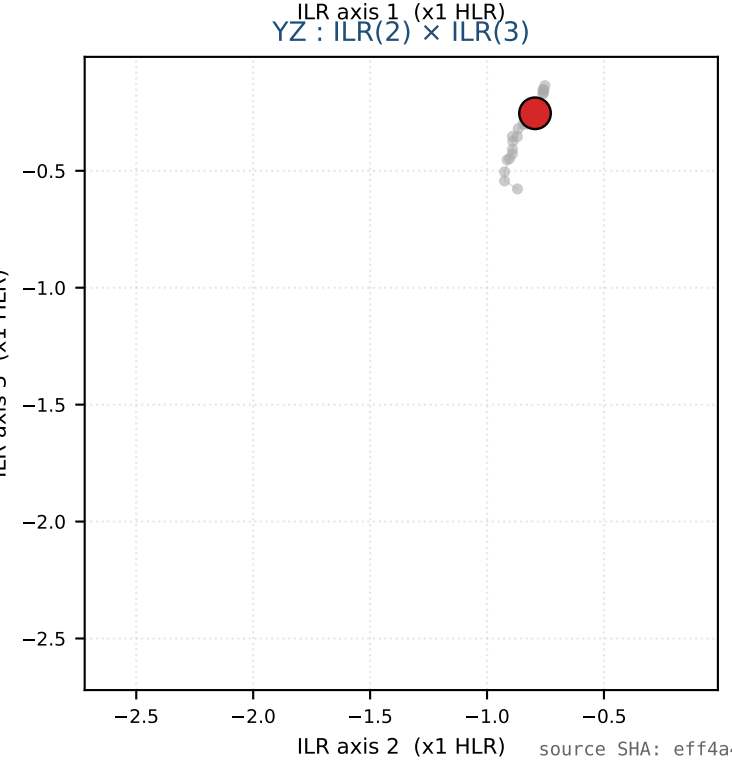
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



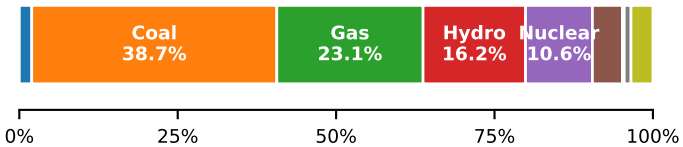




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0197	-0.911
Coal	0.3868	+2.069
Gas	0.2308	+1.553
Hydro	0.1617	+1.197
Nuclear	0.1055	+0.770
Other Fossil	0.0472	-0.034
Other Renewabl	0.0030	-2.776
Solar	0.0107	-1.522
Wind	0.0346	-0.346

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

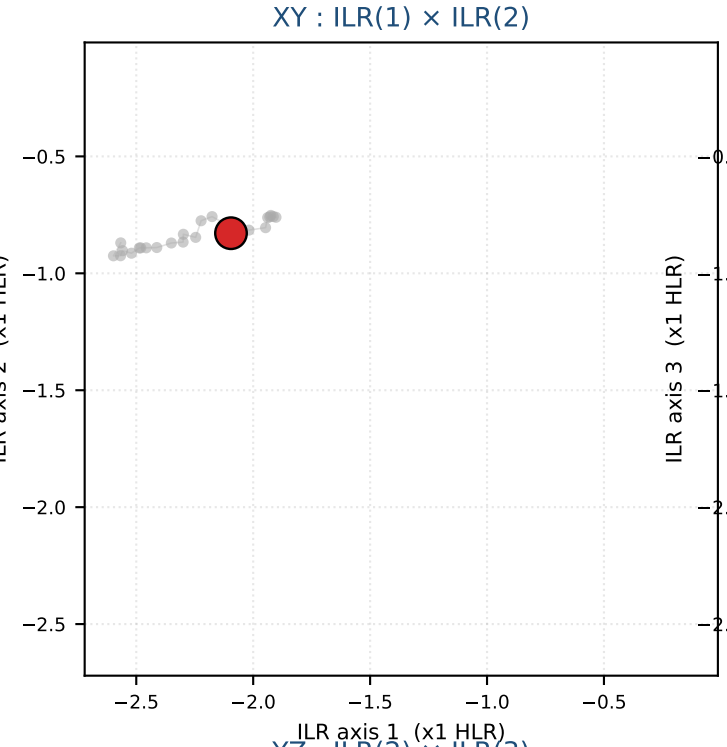
display factor: x1 (raw HLR x 1)

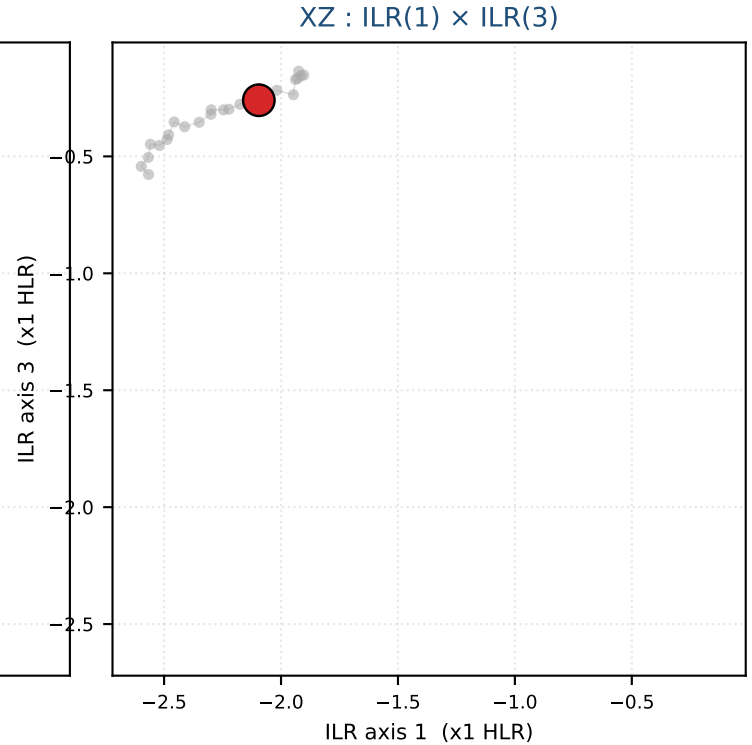
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=9 T=26 | level 1 first-principles

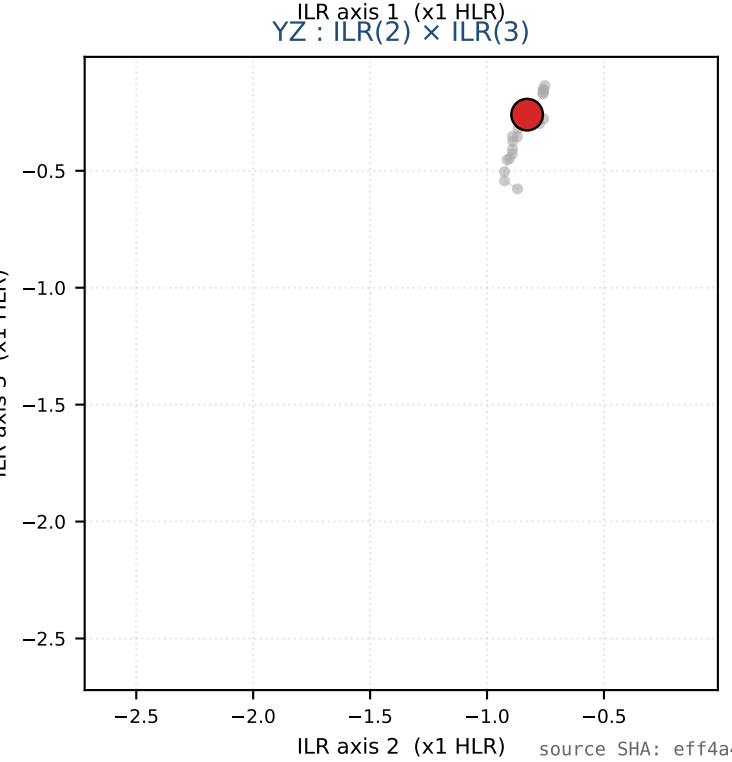
ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

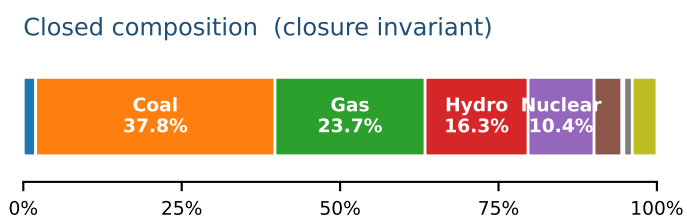




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0195	-0.946
Coal	0.3777	+2.016
Gas	0.2369	+1.550
Hydro	0.1626	+1.173
Nuclear	0.1041	+0.728
Other Fossil	0.0439	-0.135
Other Renewabl	0.0031	-2.795
Solar	0.0133	-1.333
Wind	0.0389	-0.257





closure: Σ = 1.000000 OK

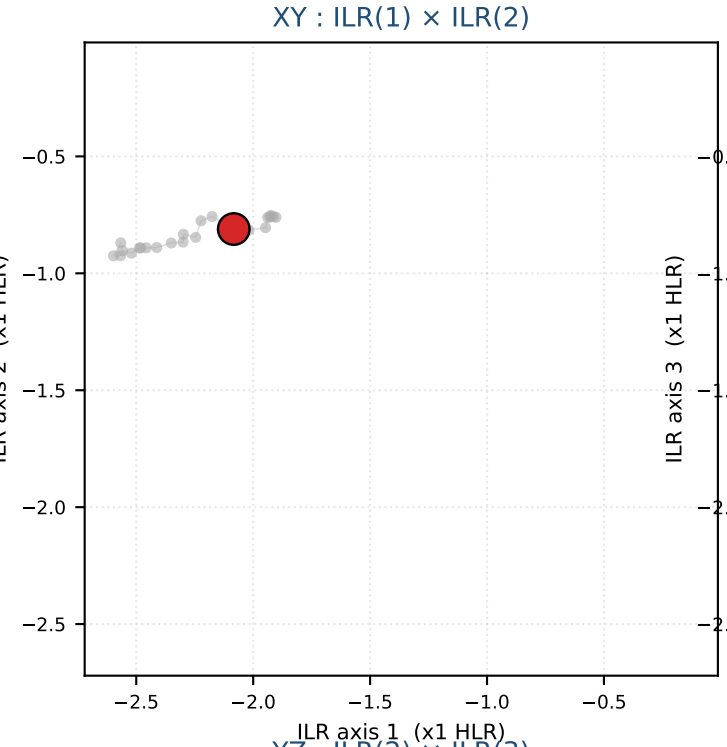
display factor: x1 (raw HLR x 1)

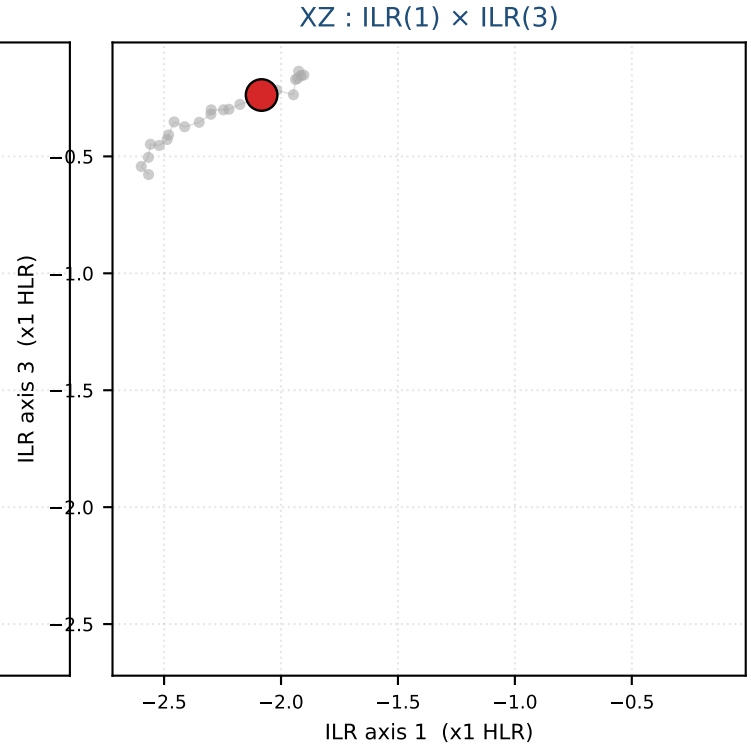
Helmert basis loadings (top 3 carriers per axis)

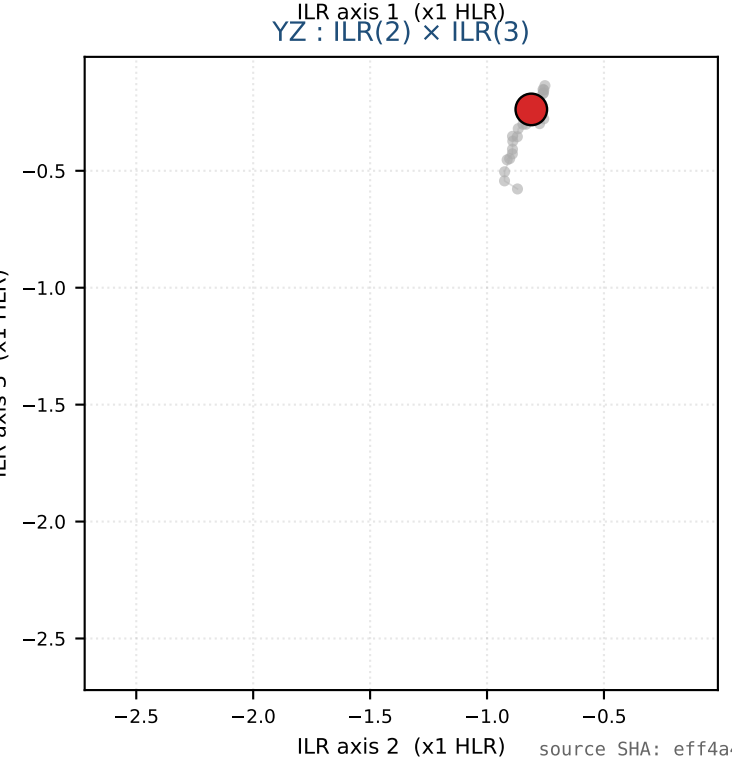
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



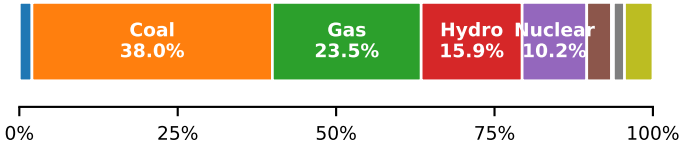




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0199	-0.957
Coal	0.3795	+1.989
Gas	0.2348	+1.509
Hydro	0.1594	+1.122
Nuclear	0.1020	+0.675
Other Fossil	0.0391	-0.284
Other Renewabl	0.0031	-2.819
Solar	0.0175	-1.089
Wind	0.0448	-0.148

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

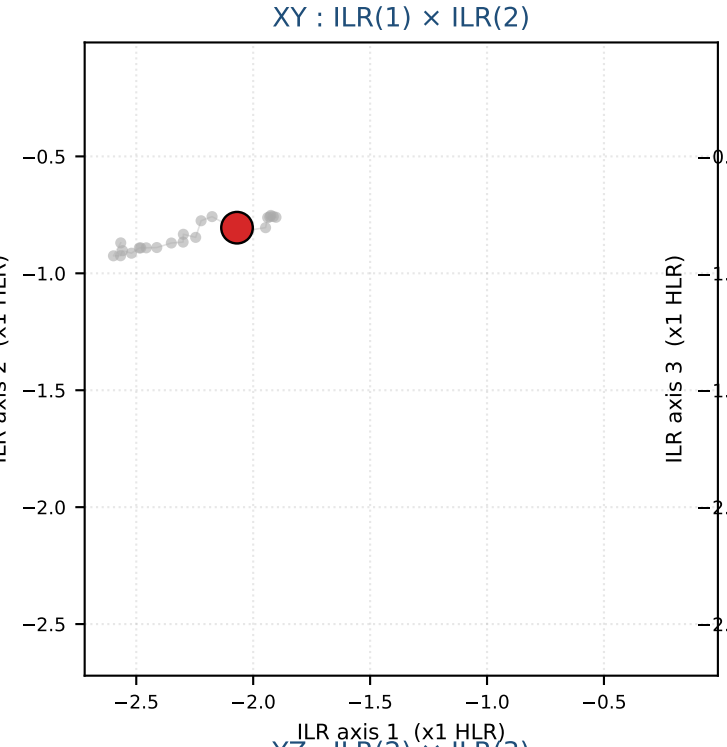
display factor: x1 (raw HLR x 1)

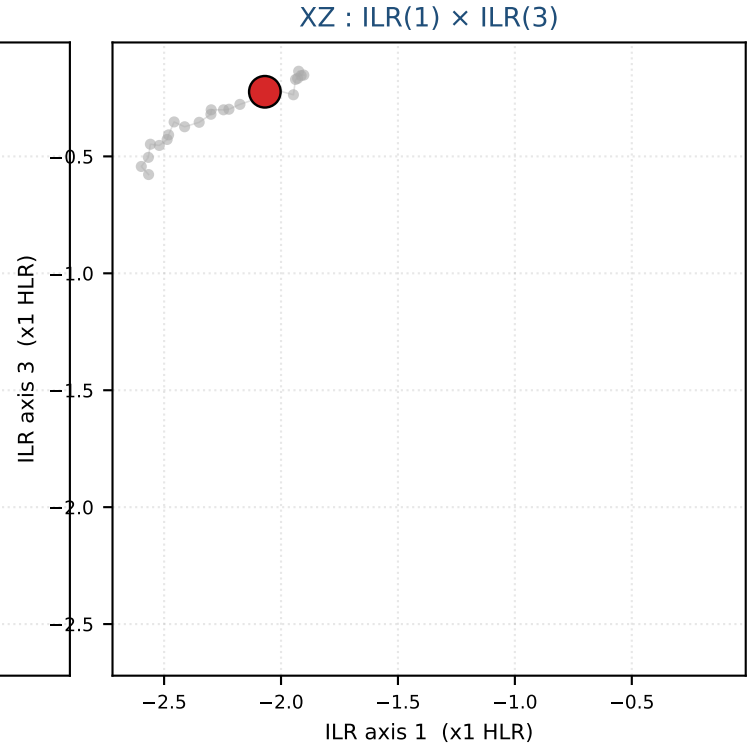
Helmert basis loadings (top 3 carriers per axis)

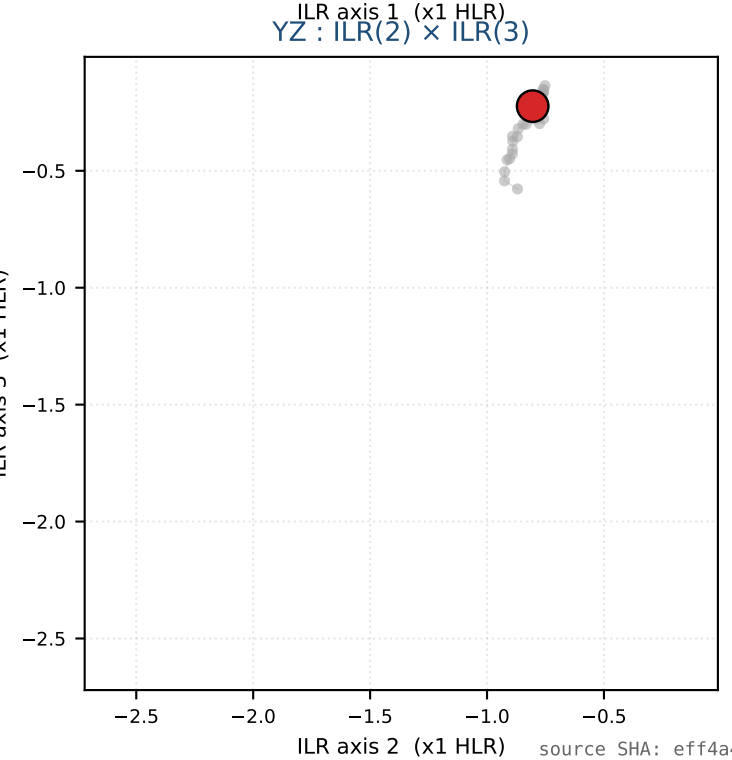
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



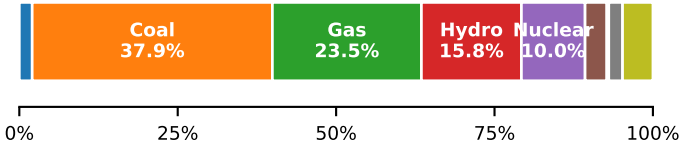




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0203	-0.954
Coal	0.3791	+1.972
Gas	0.2352	+1.495
Hydro	0.1578	+1.096
Nuclear	0.1005	+0.644
Other Fossil	0.0343	-0.430
Other Renewabl	0.0031	-2.838
Solar	0.0217	-0.887
Wind	0.0479	-0.096

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

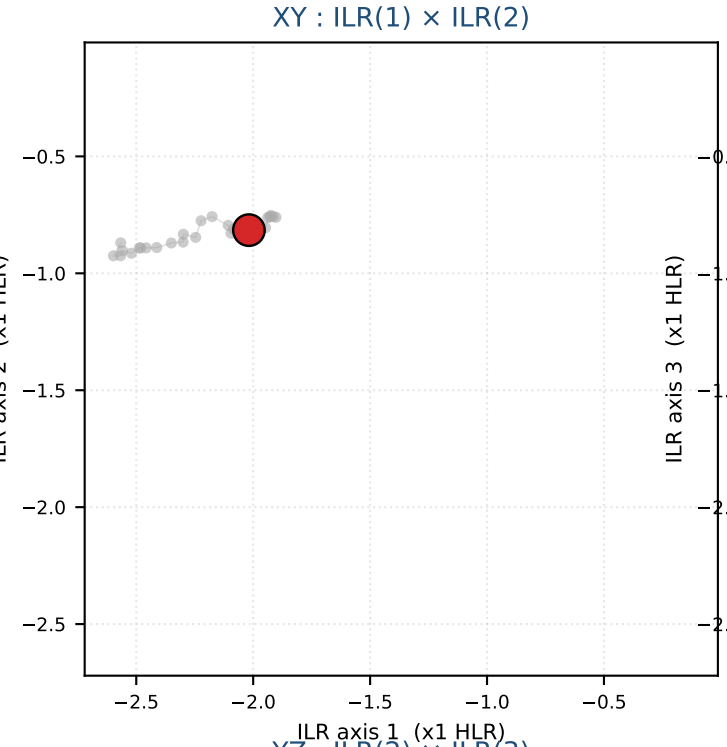
display factor: x1 (raw HLR x 1)

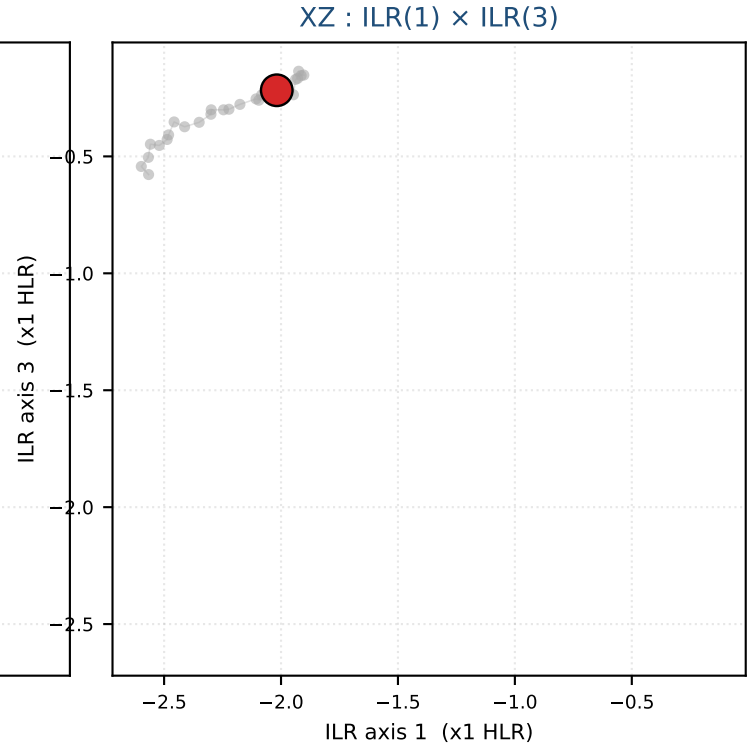
Helmert basis loadings (top 3 carriers per axis)

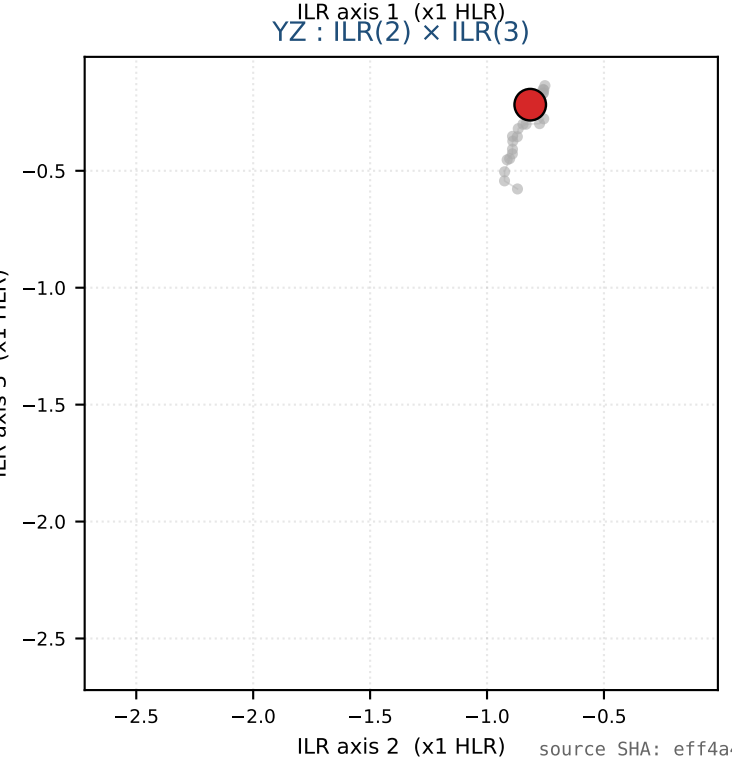
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



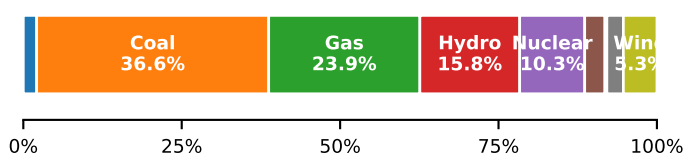




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0211	-0.944
Coal	0.3662	+1.910
Gas	0.2386	+1.482
Hydro	0.1576	+1.067
Nuclear	0.1026	+0.638
Other Fossil	0.0319	-0.531
Other Renewabl	0.0031	-2.864
Solar	0.0260	-0.734
Wind	0.0530	-0.023

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

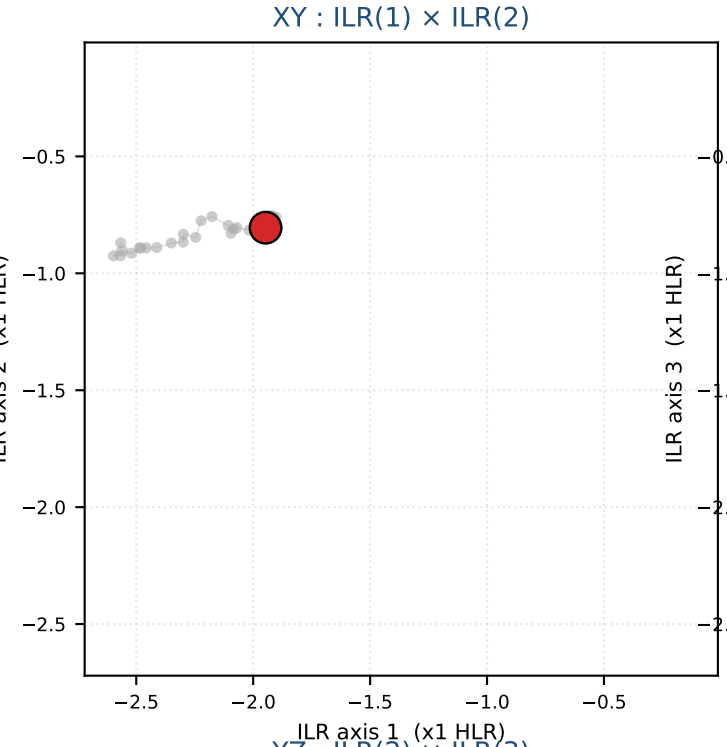
display factor: x1 (raw HLR x 1)

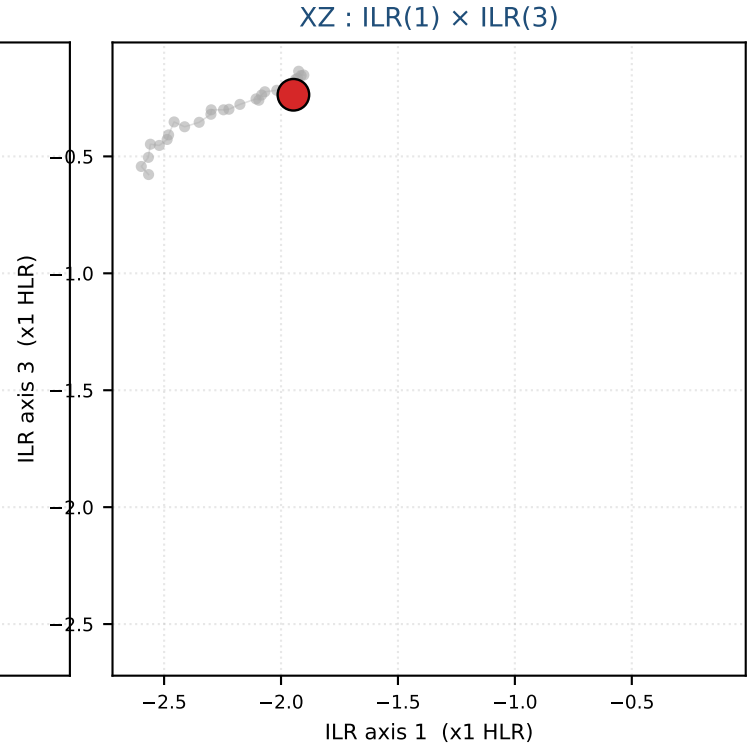
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=9 T=26 | level 1 first-principles

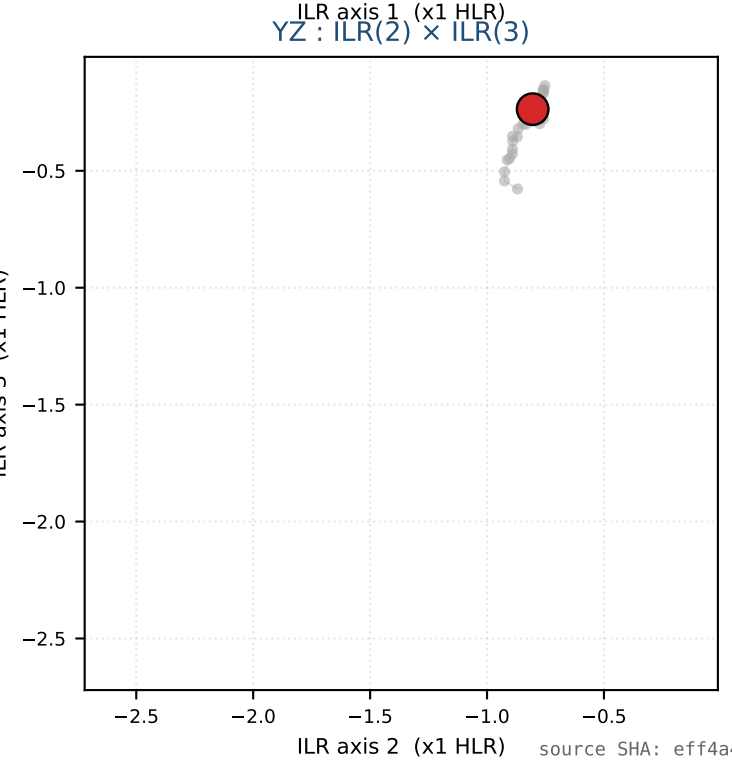
ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

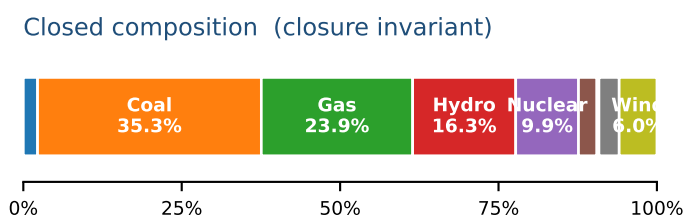




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0225	-0.913
Coal	0.3531	+1.841
Gas	0.2389	+1.450
Hydro	0.1627	+1.066
Nuclear	0.0991	+0.570
Other Fossil	0.0290	-0.657
Other Renewabl	0.0032	-2.856
Solar	0.0319	-0.562
Wind	0.0596	+0.061





closure: Σ = 1.000000 OK

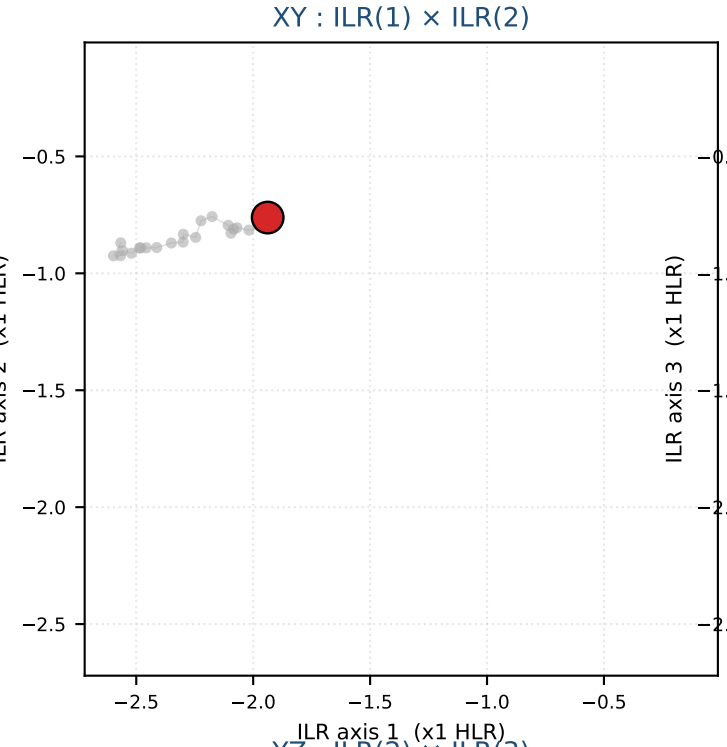
display factor: x1 (raw HLR x 1)

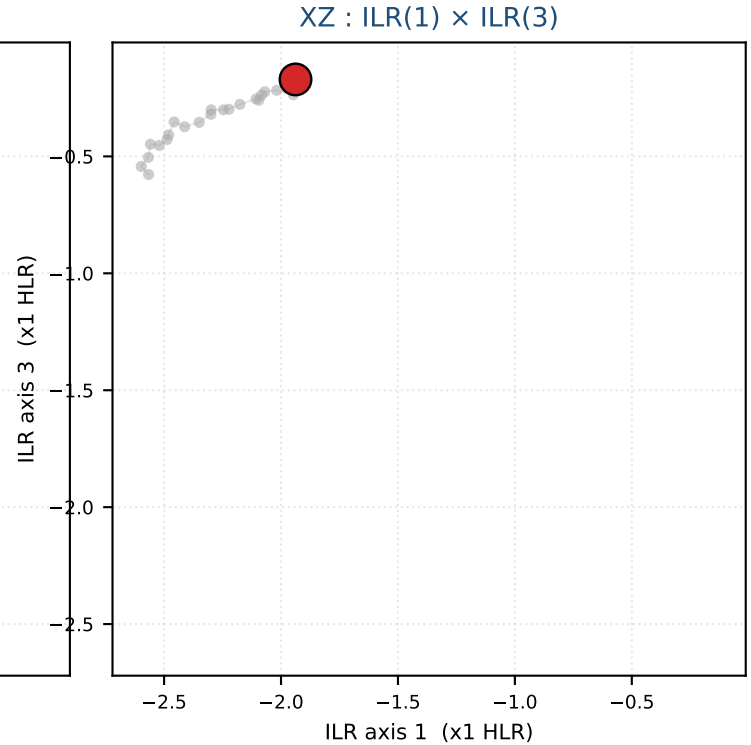
Helmert basis loadings (top 3 carriers per axis)

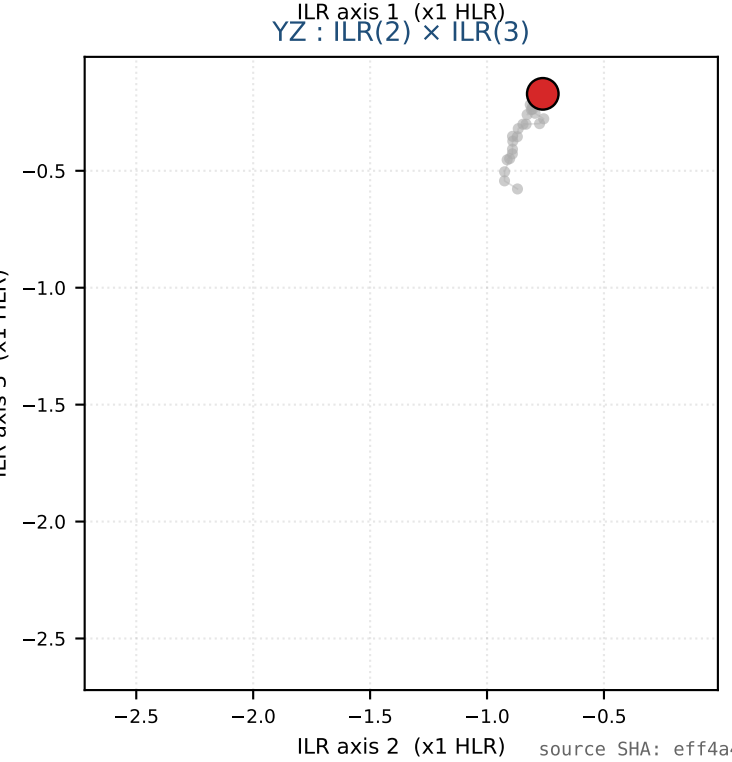
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



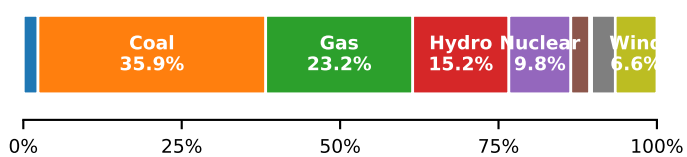




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0232	-0.900
Coal	0.3594	+1.841
Gas	0.2320	+1.403
Hydro	0.1518	+0.978
Nuclear	0.0978	+0.539
Other Fossil	0.0298	-0.649
Other Renewabl	0.0031	-2.927
Solar	0.0373	-0.425
Wind	0.0657	+0.141

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

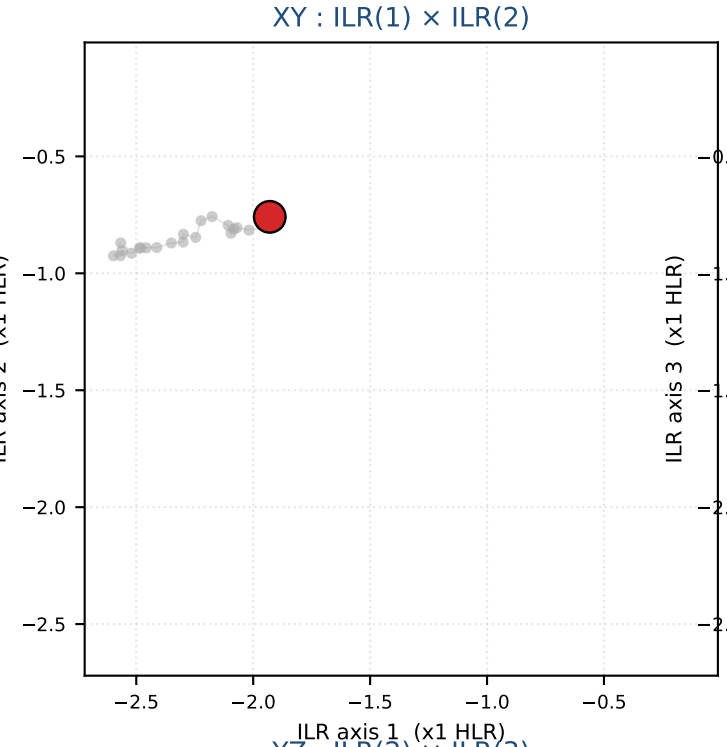
display factor: x1 (raw HLR x 1)

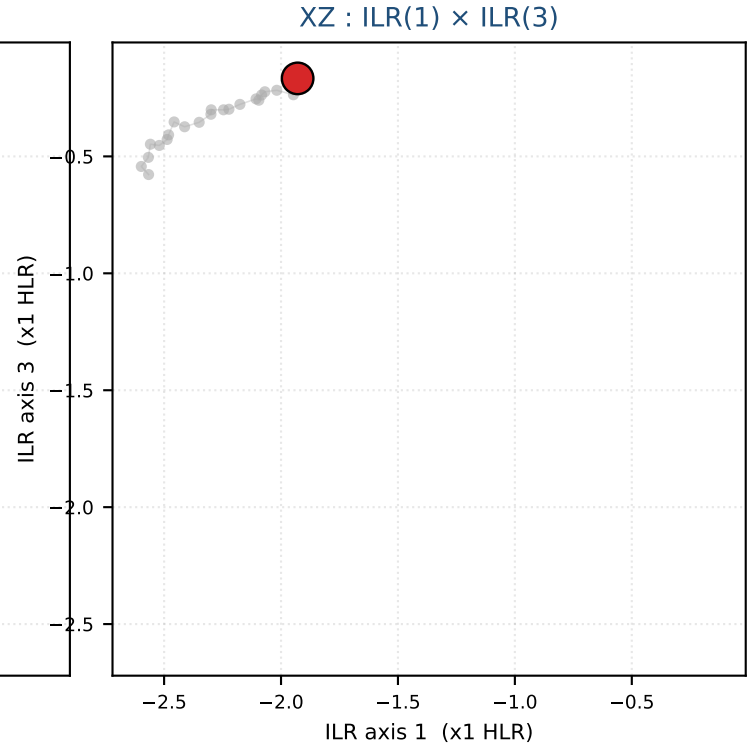
Helmert basis loadings (top 3 carriers per axis)

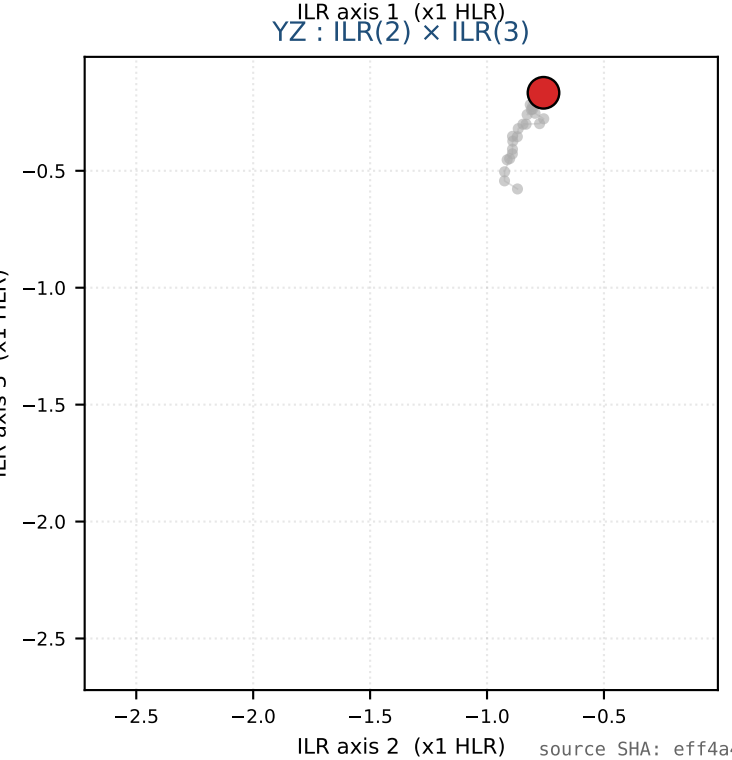
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

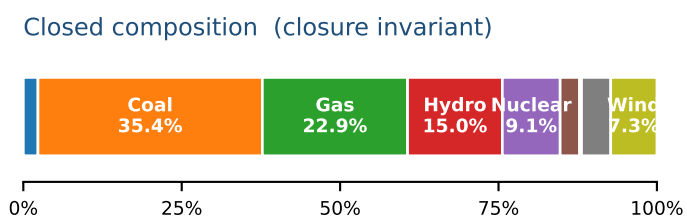






Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0231	-0.928
Coal	0.3541	+1.800
Gas	0.2292	+1.365
Hydro	0.1495	+0.938
Nuclear	0.0913	+0.445
Other Fossil	0.0307	-0.645
Other Renewabl	0.0030	-2.957
Solar	0.0461	-0.238
Wind	0.0729	+0.220



closure: Σ = 1.000000 OK

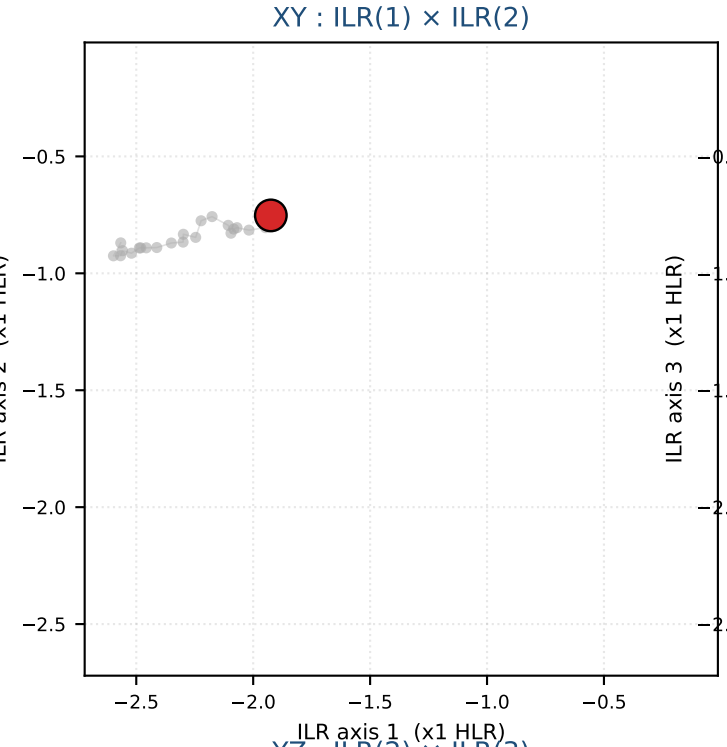
display factor: x1 (raw HLR x 1)

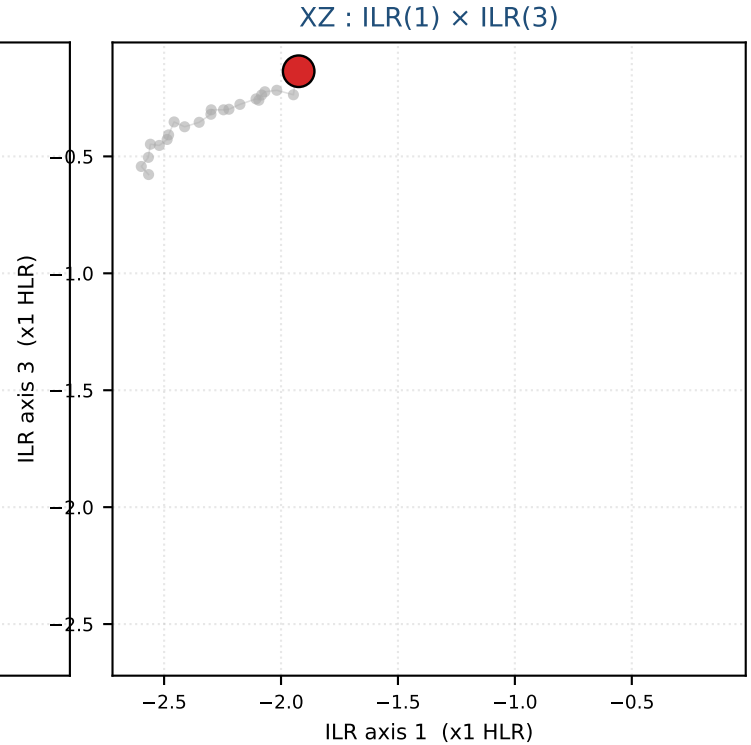
Helmert basis loadings (top 3 carriers per axis)

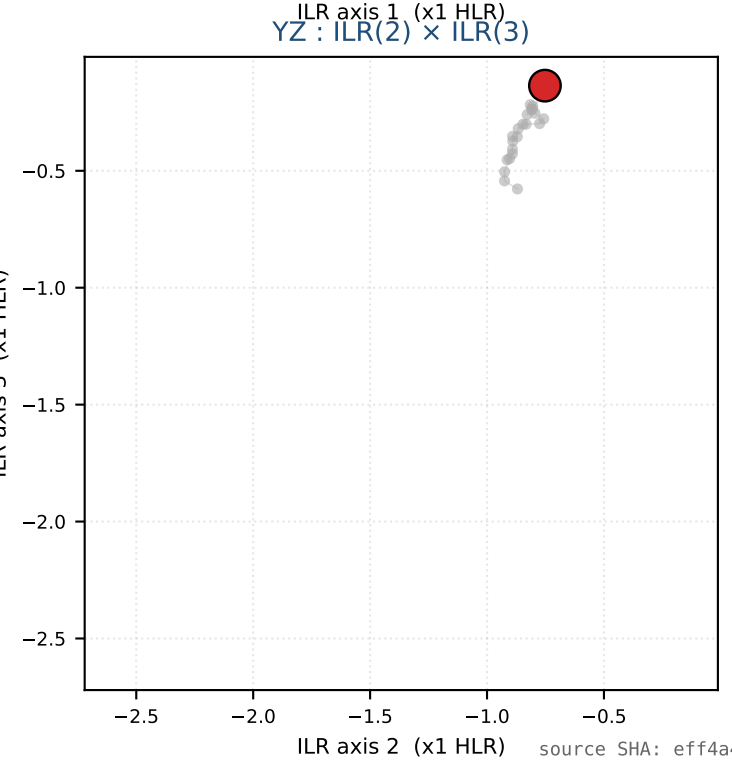
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



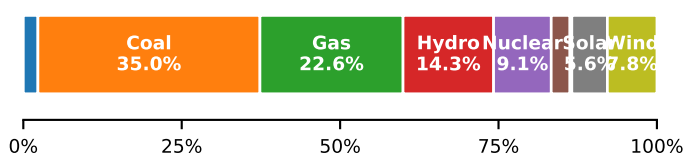




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0230	-0.946
Coal	0.3504	+1.776
Gas	0.2258	+1.336
Hydro	0.1429	+0.879
Nuclear	0.0909	+0.426
Other Fossil	0.0296	-0.697
Other Renewabl	0.0030	-2.992
Solar	0.0561	-0.057
Wind	0.0782	+0.275

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

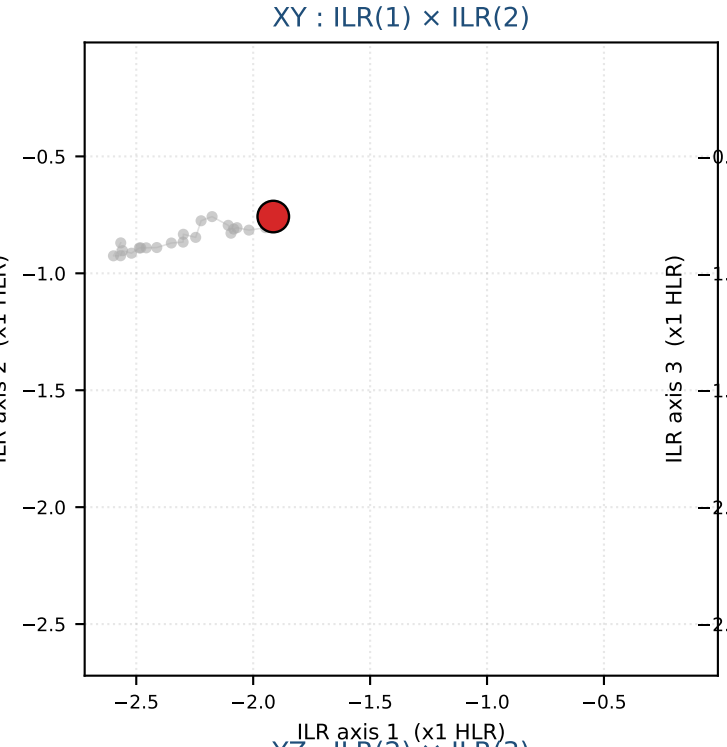
display factor: x1 (raw HLR x 1)

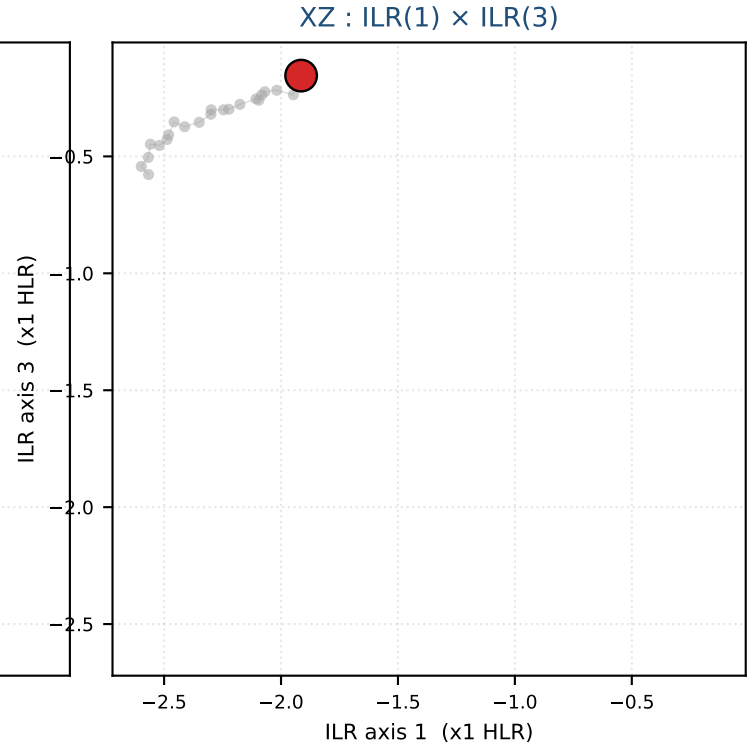
Helmert basis loadings (top 3 carriers per axis)

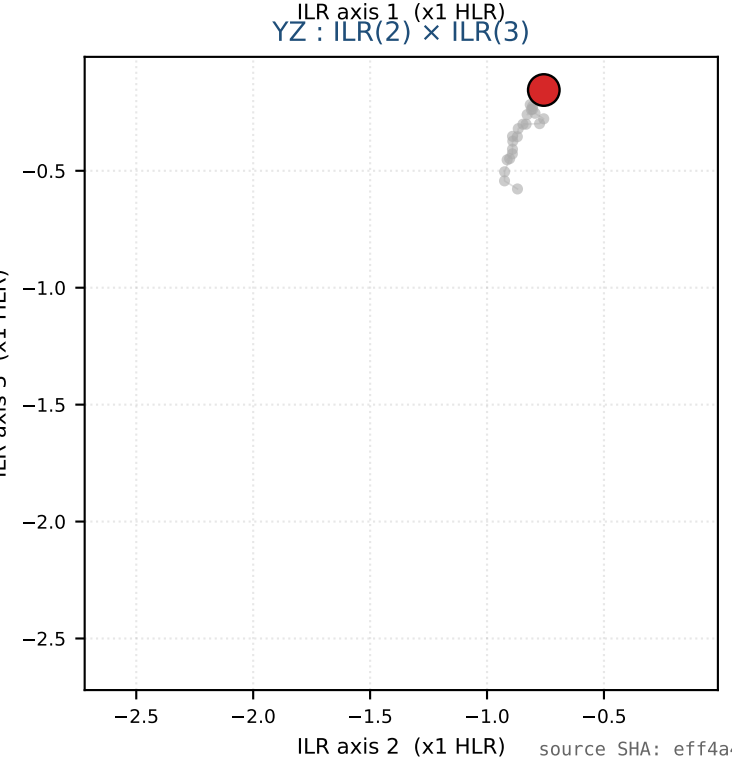
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



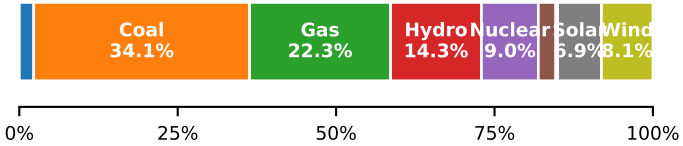




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0227	-0.968
Coal	0.3407	+1.739
Gas	0.2225	+1.313
Hydro	0.1433	+0.873
Nuclear	0.0898	+0.406
Other Fossil	0.0276	-0.774
Other Renewabl	0.0029	-3.041
Solar	0.0693	+0.146
Wind	0.0811	+0.304

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

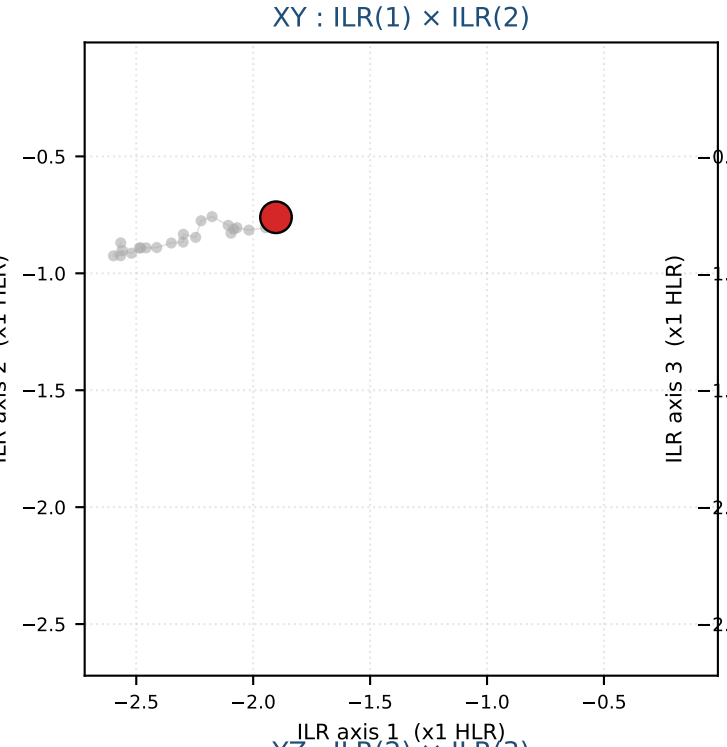
display factor: x1 (raw HLR x 1)

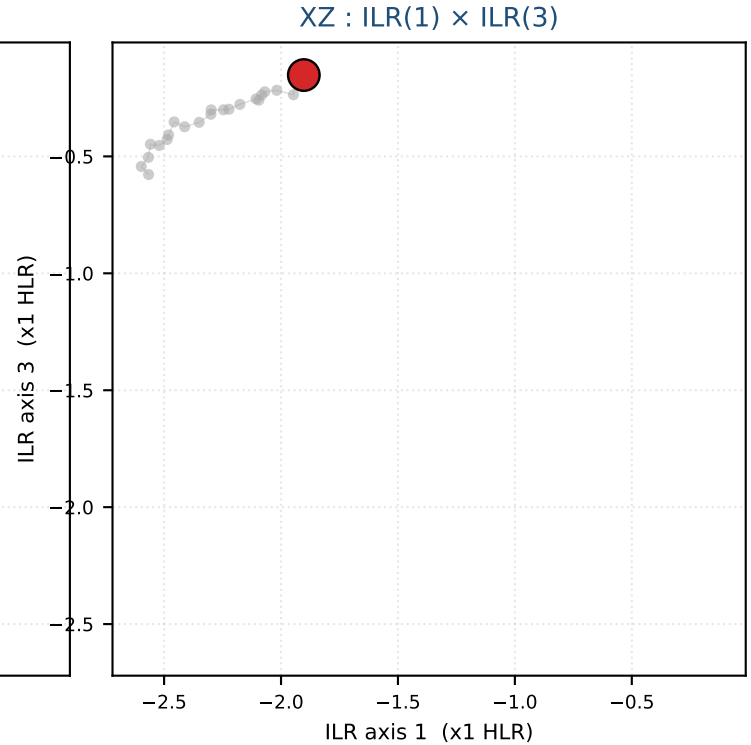
Helmert basis loadings (top 3 carriers per axis)

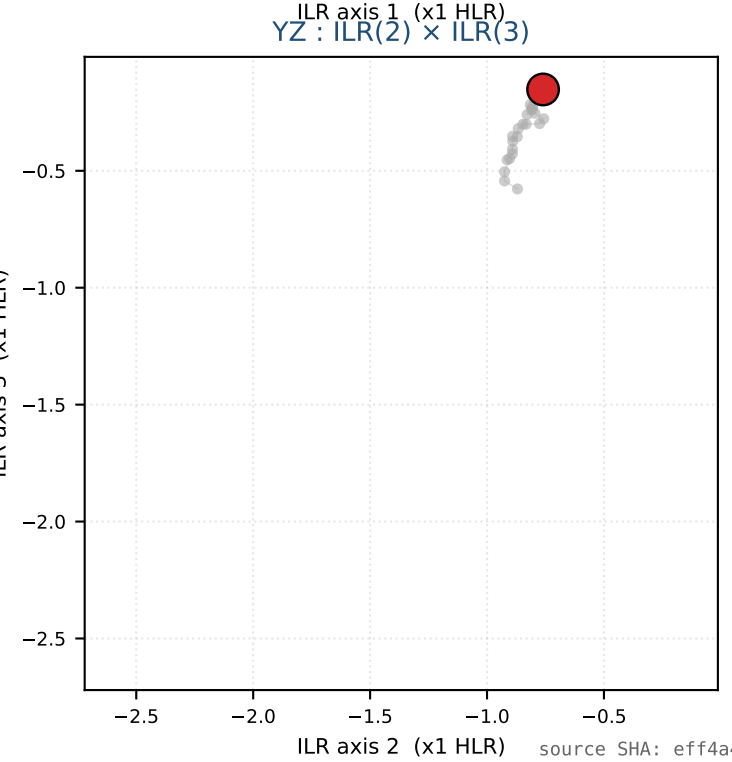
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=9 T=26 | level 1 first-principles

ember_wld | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



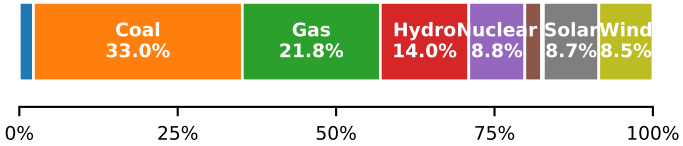




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0224	-0.997
Coal	0.3296	+1.693
Gas	0.2178	+1.279
Hydro	0.1396	+0.834
Nuclear	0.0885	+0.379
Other Fossil	0.0265	-0.828
Other Renewabl	0.0028	-3.071
Solar	0.0875	+0.367
Wind	0.0854	+0.343

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

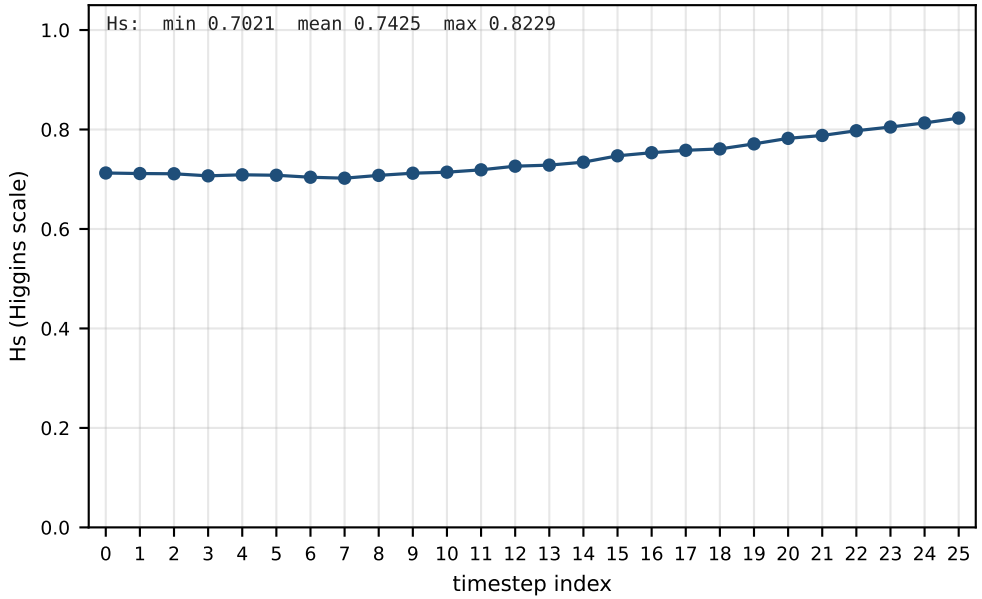
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

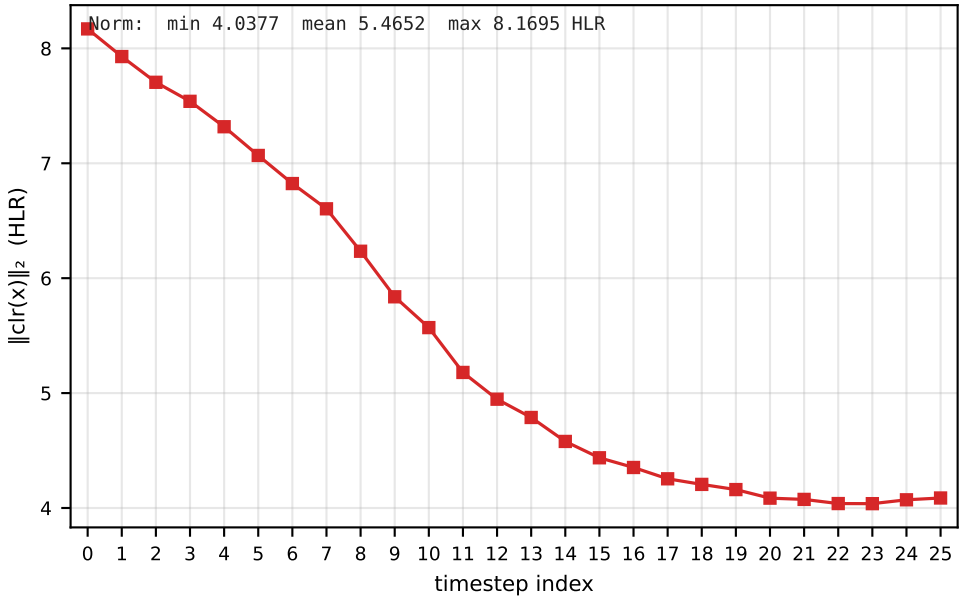
STAGE 1 SUMMARY | ember_wld | D=9 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.3 | Schema 2.0.0 | Generated 2026-05-05T20:04:43

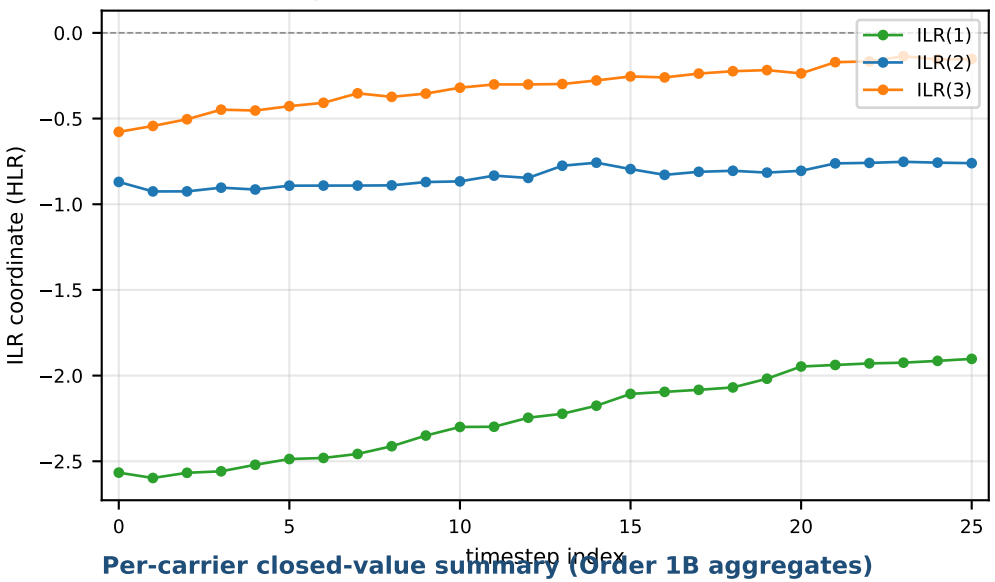
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 4.44e-16
closure status        :  OK ✓

Hs range              :  [0.7021, 0.8229]
Aitchison norm range  :  [4.0377, 8.1695] HLR
ILR(1) range          :  [-2.5976, -1.9025]
ILR(2) range          :  [-0.9252, -0.7524]
ILR(3) range          :  [-0.5777, -0.1359]

Source file           :  ember_WLD_World_generation_TWh.csv
Source SHA-256        :  eff4a4d3ffe97ec12432bdda954ee422...
Closed-data SHA-256   :  fc39ef301f19a2972ca8efc2a28b9373...
Content SHA-256       :  9ee6475abb1ecdf05726bee6f0798d38...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0096	0.0168	0.0232	0.0047
1	Coal	0.3296	0.3817	0.4084	0.0217
2	Gas	0.1797	0.2183	0.2389	0.0160
3	Hydro	0.1396	0.1583	0.1721	0.0073
4	Nuclear	0.0885	0.1212	0.1686	0.0264
5	Other Fossil	0.0265	0.0513	0.0867	0.0184
6	Other Renewabl	0.0027	0.0029	0.0032	0.0001
7	Solar	0.0001	0.0170	0.0875	0.0236
8	Wind	0.0020	0.0324	0.0854	0.0271